AdaBoost

Machine Learning Techniques

Karthik Thiagarajan

AdaBoost

<u>Input</u>

- $D = \{(\mathbf{x}_1, y_1), \cdots, (\mathbf{x}_n, y_n)\}$
  - $y_i \in \{1, -1\}$
- weak-learner (decision stump)
  - $h : \mathbb{R}^d \to \{1, -1\}$
- T, number of rounds

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<u>Initialization</u>

$D_0(i) = \dfrac{1}{n}$, for all $1 \leqslant i \leqslant n$

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for t in $[1, \cdots, T]$:

$$h_t \leftarrow \text{weak-learner}(D, D_{t-1})$$

$$\epsilon_t = \text{error-rate}(h_t)$$

$$\alpha_t = \frac{1}{2} \ln \frac{1 - \epsilon_t}{\epsilon_t}$$

$$D_t(i) = \begin{cases} e^{\alpha_t} \cdot D_{t-1}(i), & h_t(x_i) \neq y_i \\ e^{-\alpha_t} \cdot D_{t-1}(i), & h_t(x_i) = y_i \end{cases}, \quad 1 \leqslant i \leqslant n$$

$$D_t(i) = \frac{D_t(i)}{\sum_{i=1}^{n} D_t(i)}$$

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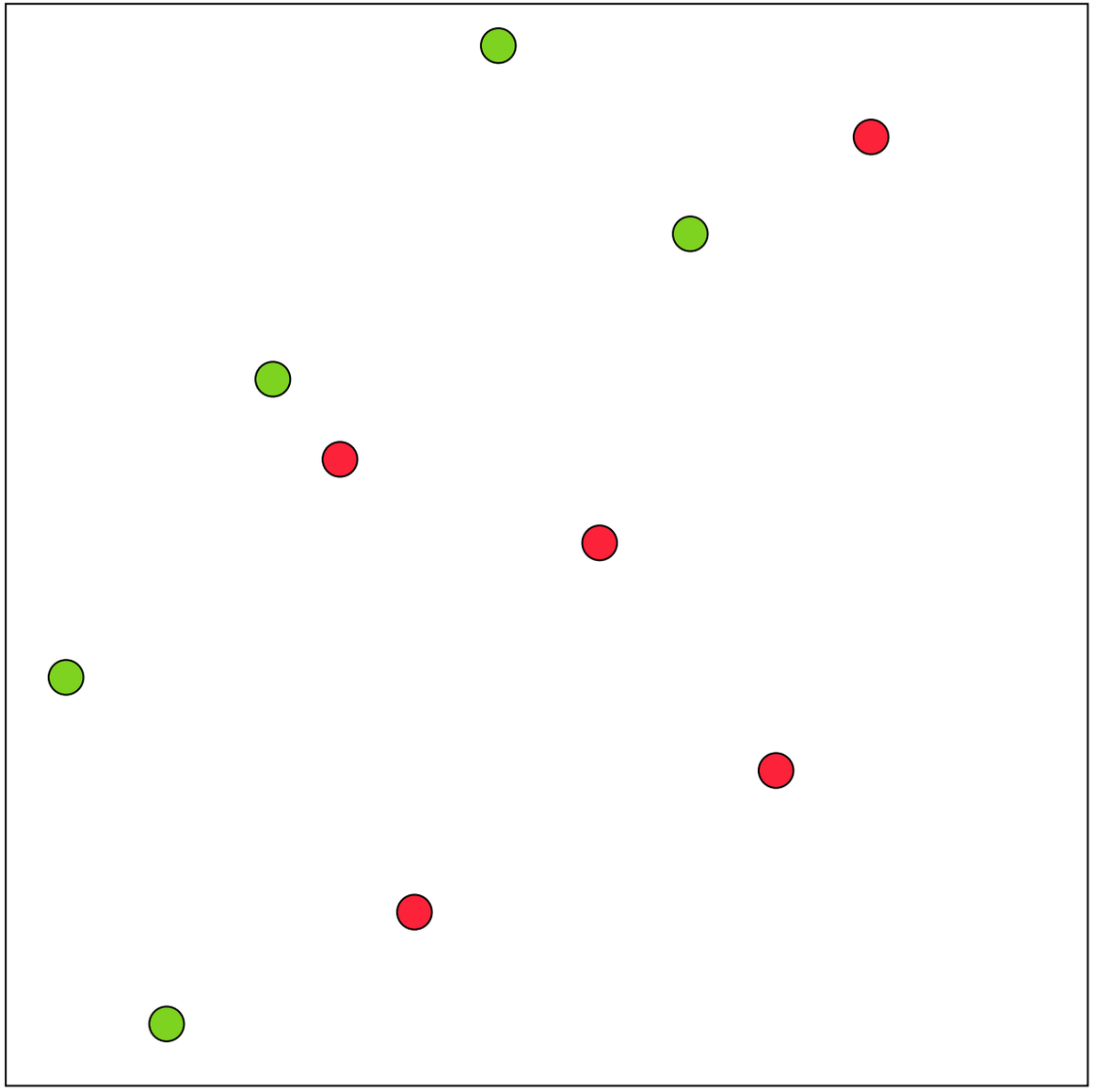
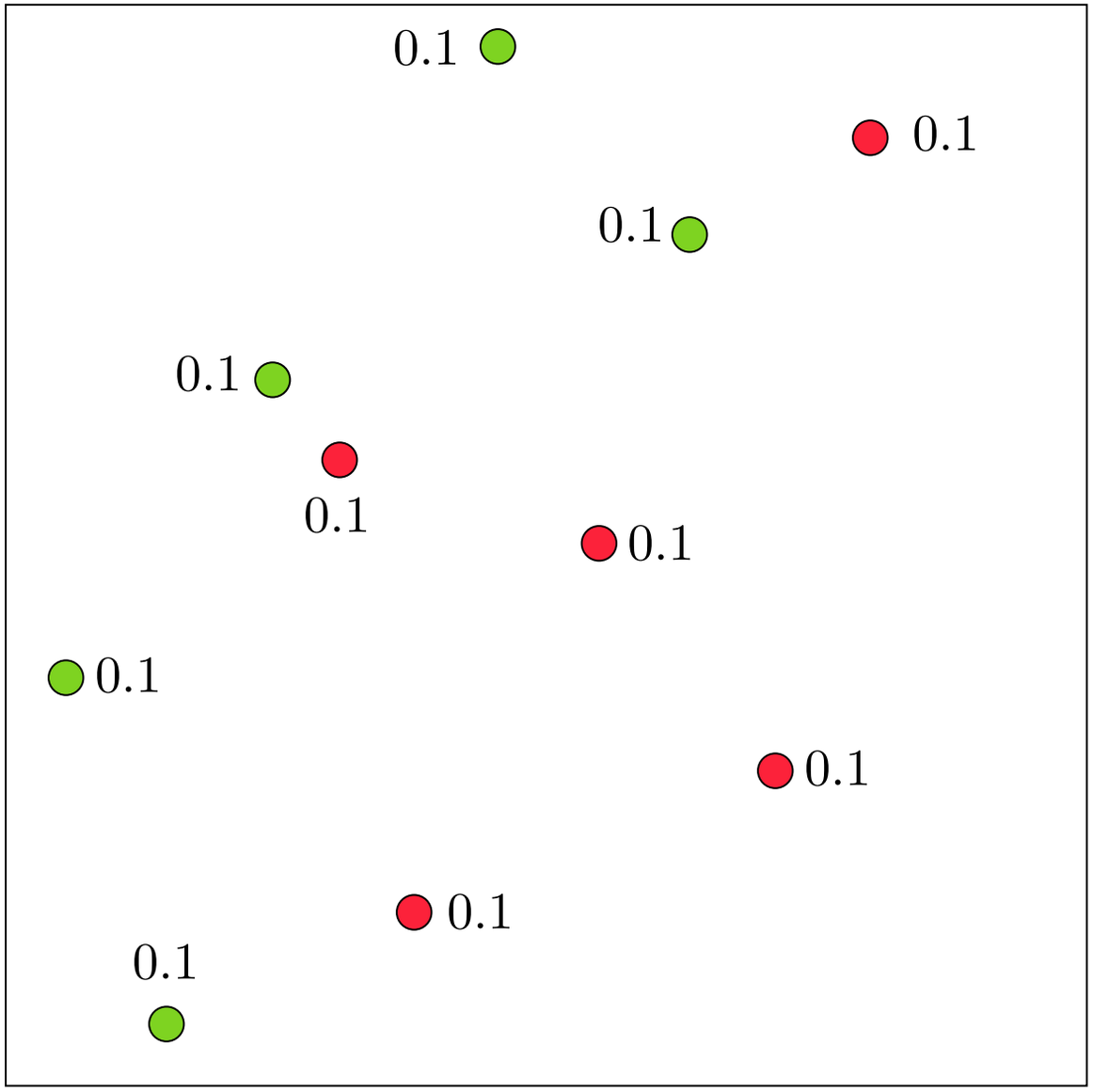
Output

return $h(\,\cdot\,) = \text{sign}\left[\sum_{t=1}^{T} \alpha_t h_t(\,\cdot\,)\right]$

AdaBoost | Example

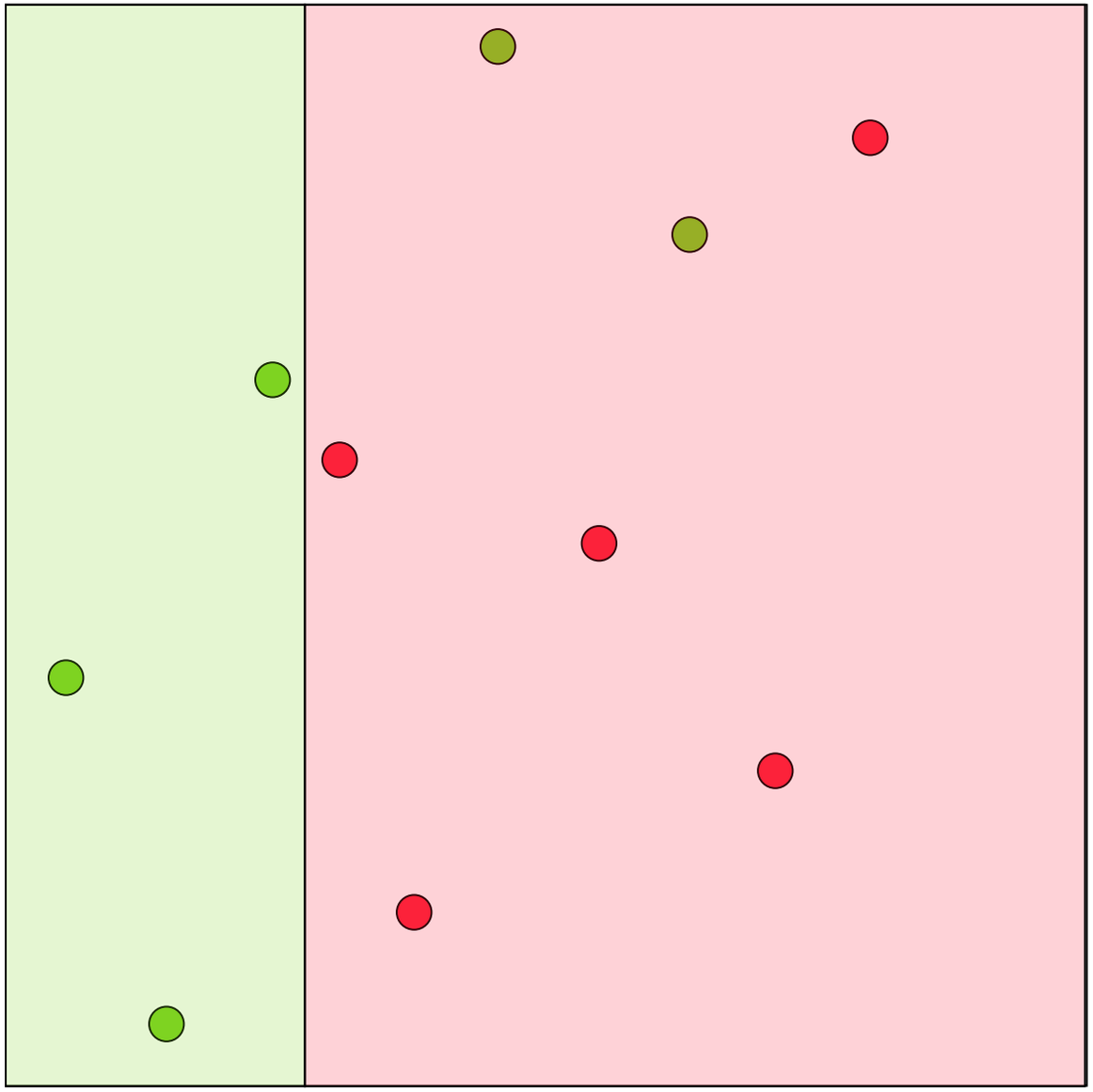


AdaBoost | Round-0

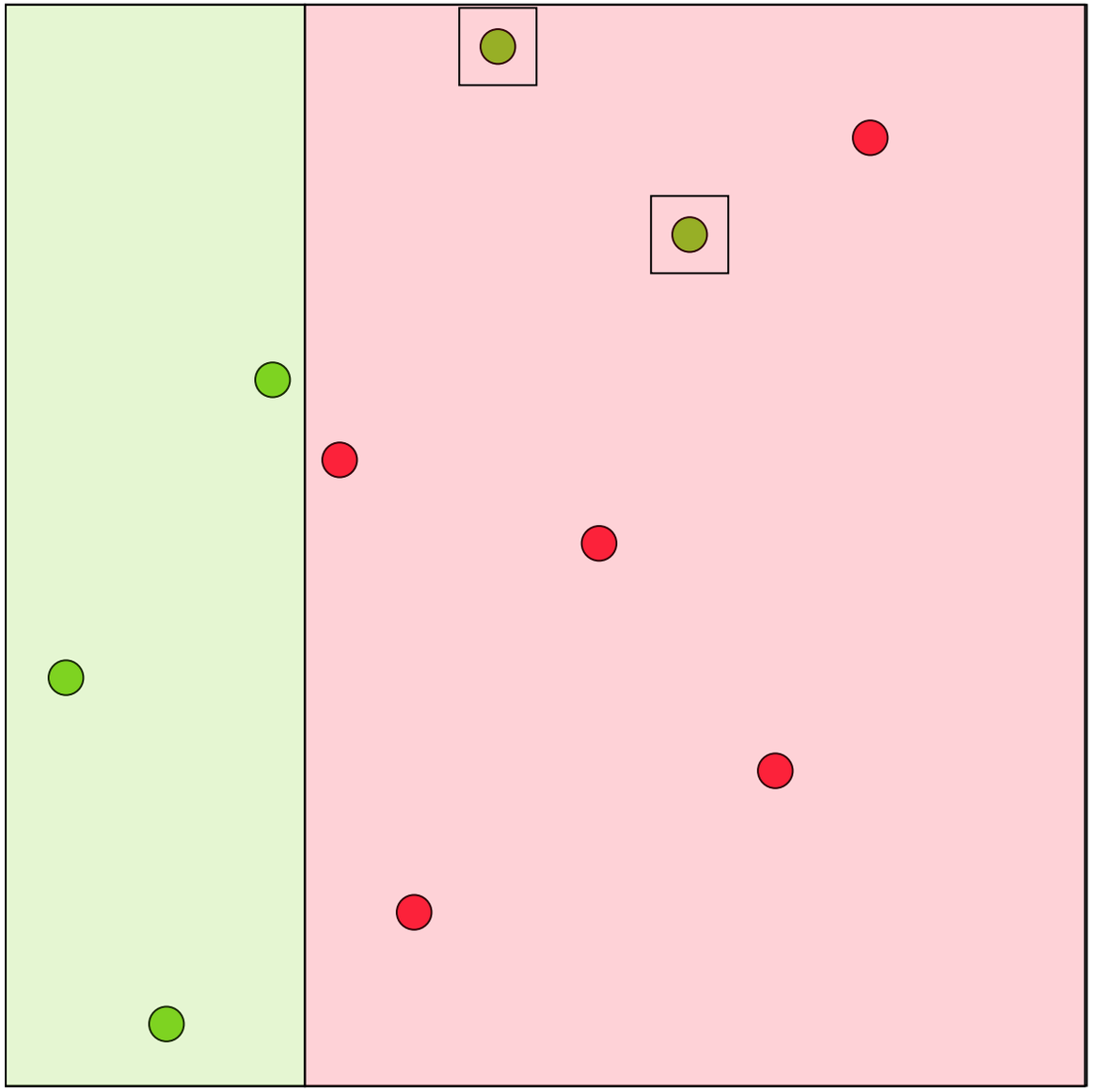


$$D_0(i) = \frac{1}{n} = \frac{1}{10} = 0.1$$

AdaBoost | Round-1

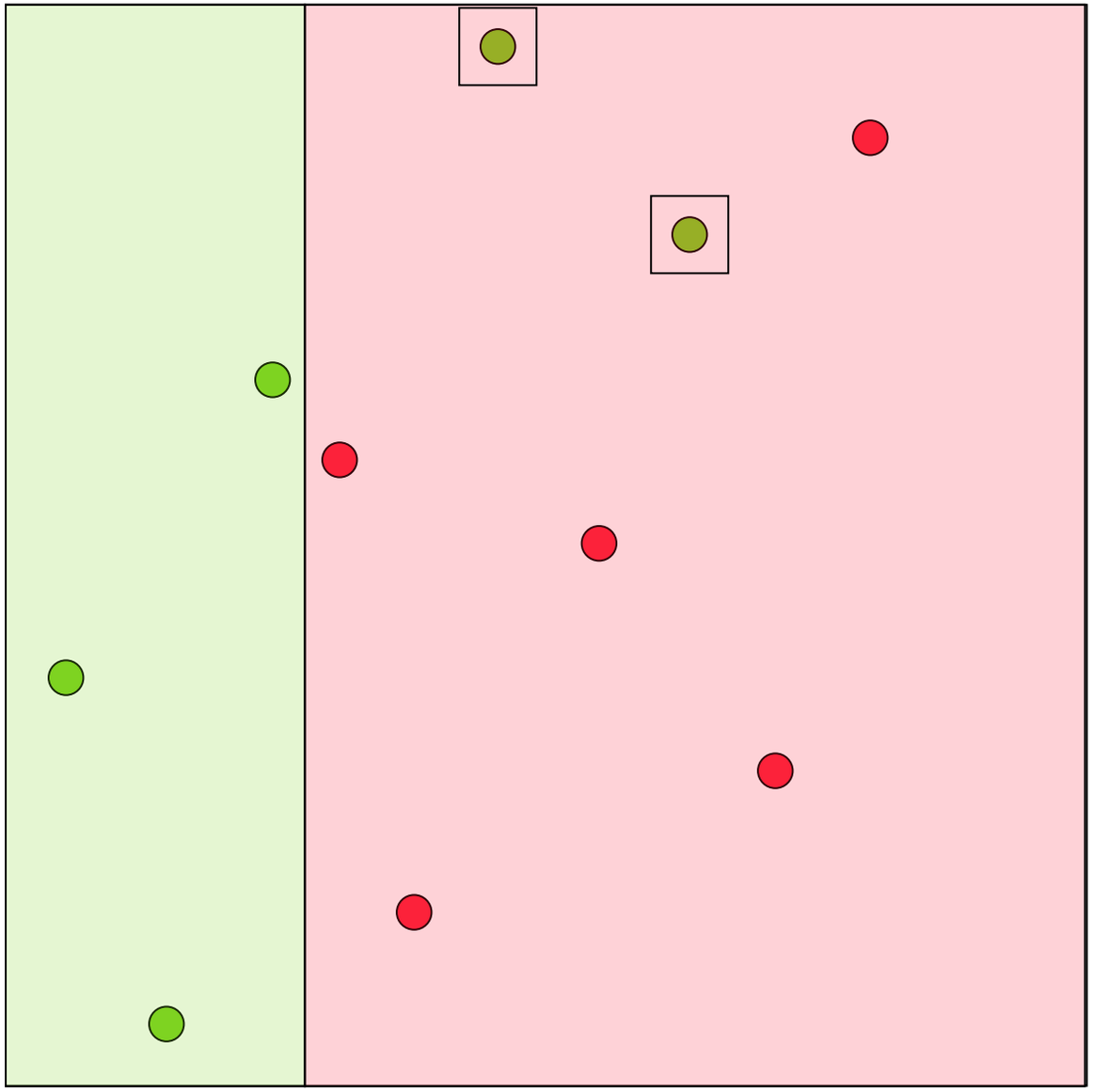


AdaBoost | Round-1



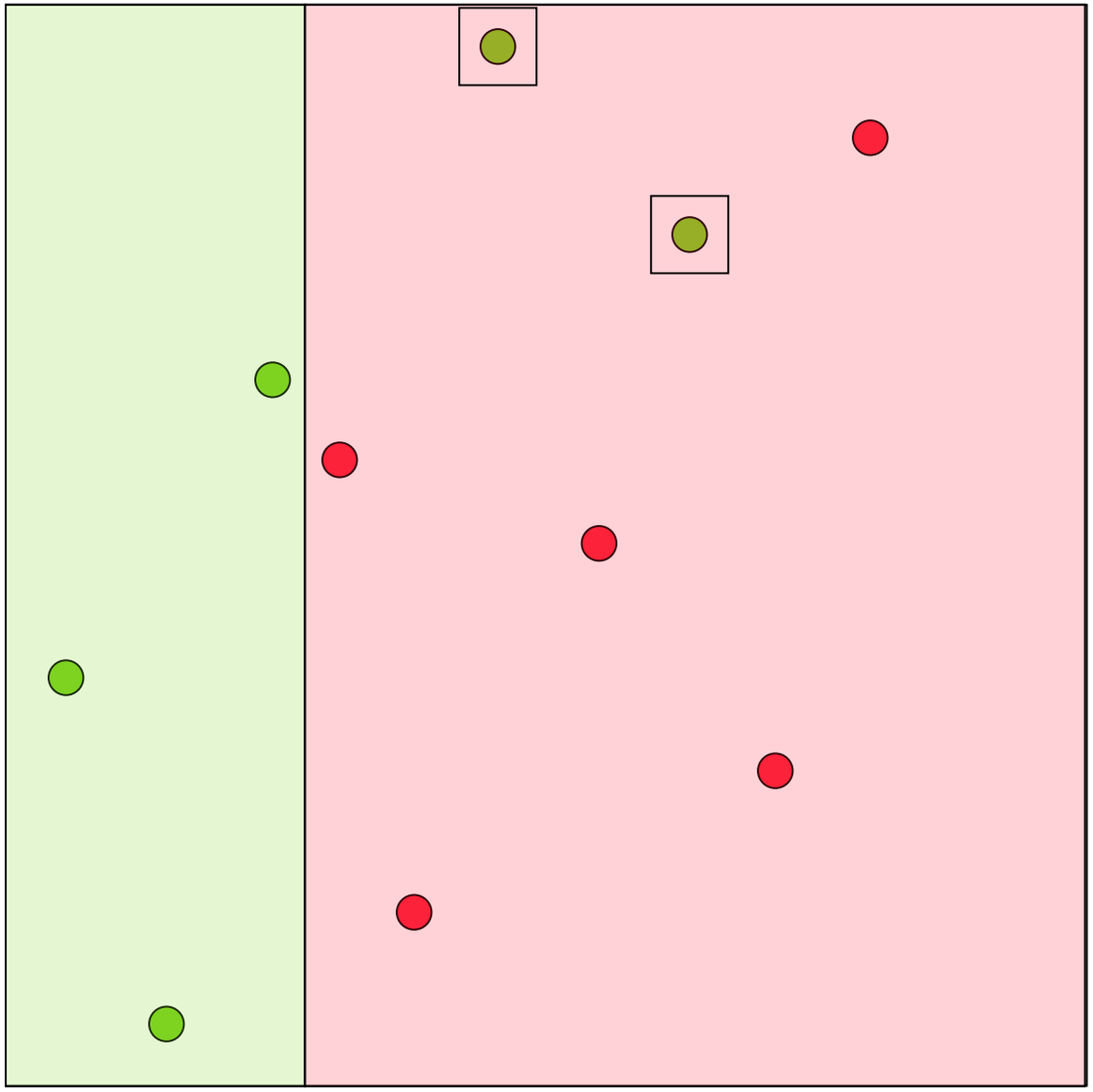
$$\epsilon_1 = 0.1 + 0.1 = 0.2$$

AdaBoost | Round-1



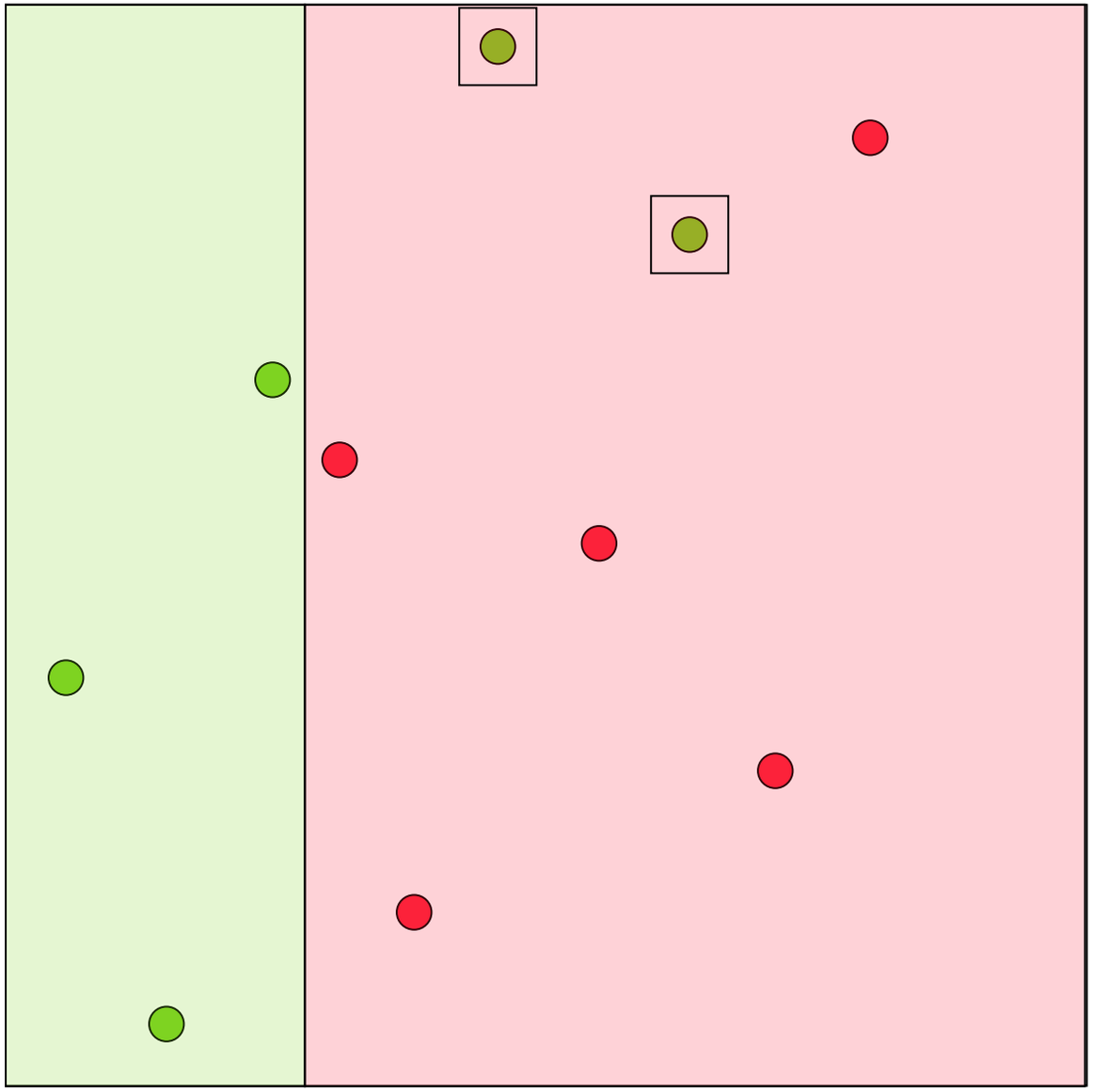
$$\alpha_1 = \frac{1}{2} \ln\left(\frac{1-\epsilon_1}{\epsilon_1}\right)$$

AdaBoost | Round-1



$$\alpha_1 = \frac{1}{2}\ln\left(\frac{1-\epsilon_1}{\epsilon_1}\right) = \frac{1}{2}\ln\left(\frac{1-0.2}{0.2}\right) \approx 0.69$$

AdaBoost | Round-1



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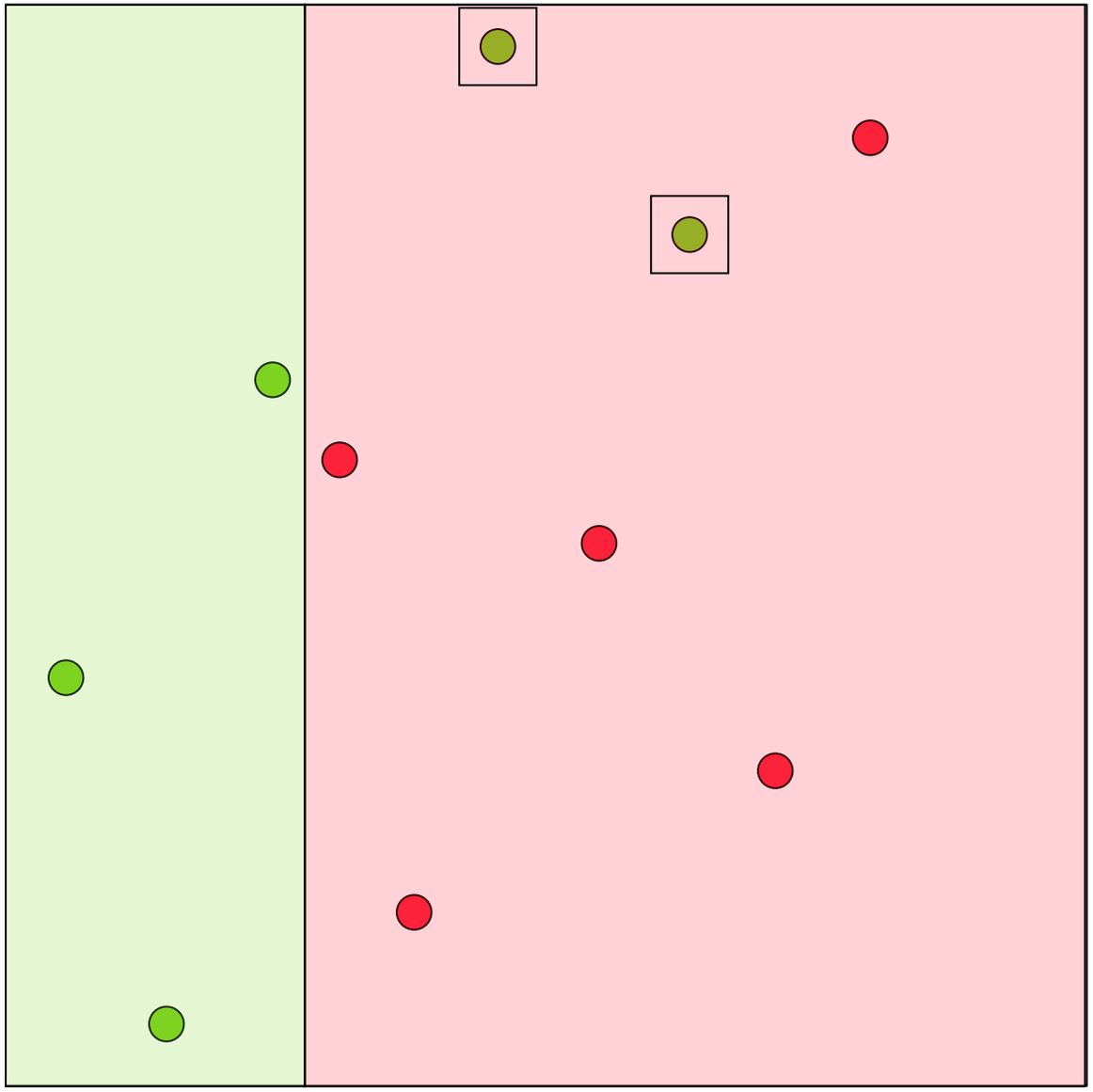
$$\widetilde{D}_1(i) = \begin{cases} D_0(i)\cdot e^{\alpha_1}, & \text{mistake} \\ D_0(i)\cdot e^{-\alpha_1}, & \text{not a mistake} \end{cases}$$

AdaBoost | Round-1



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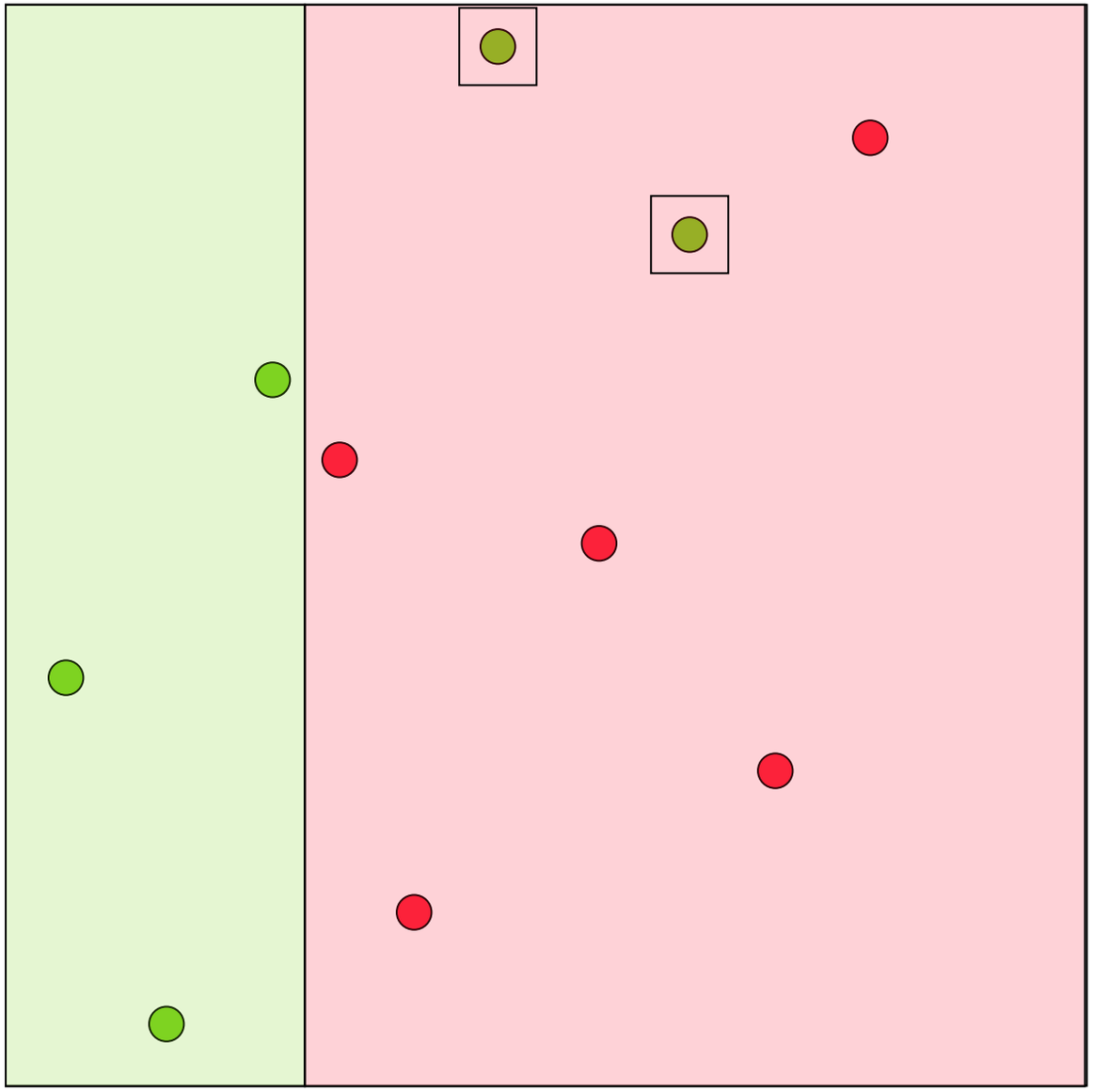
$$\widetilde{D}_1(i) = \begin{cases} D_0(i)\times 2, & \text{mistake} \\ D_0(i)\times 0.5, & \text{not a mistake} \end{cases}$$

$$\widetilde{D}_1(i) = \begin{cases} 0.2, & \text{mistake} \\ 0.05, & \text{not a mistake} \end{cases}$$

AdaBoost | Round-1



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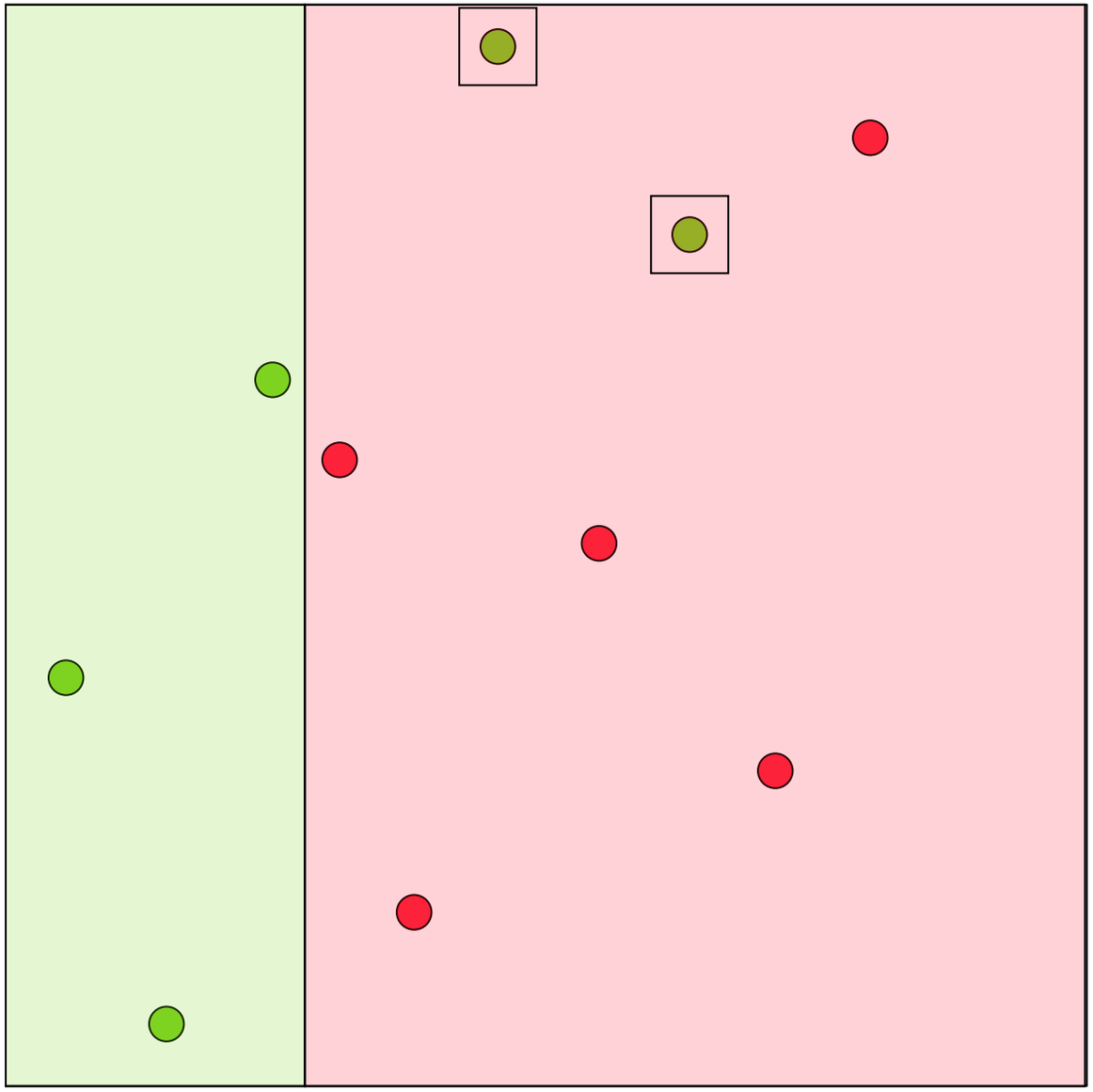
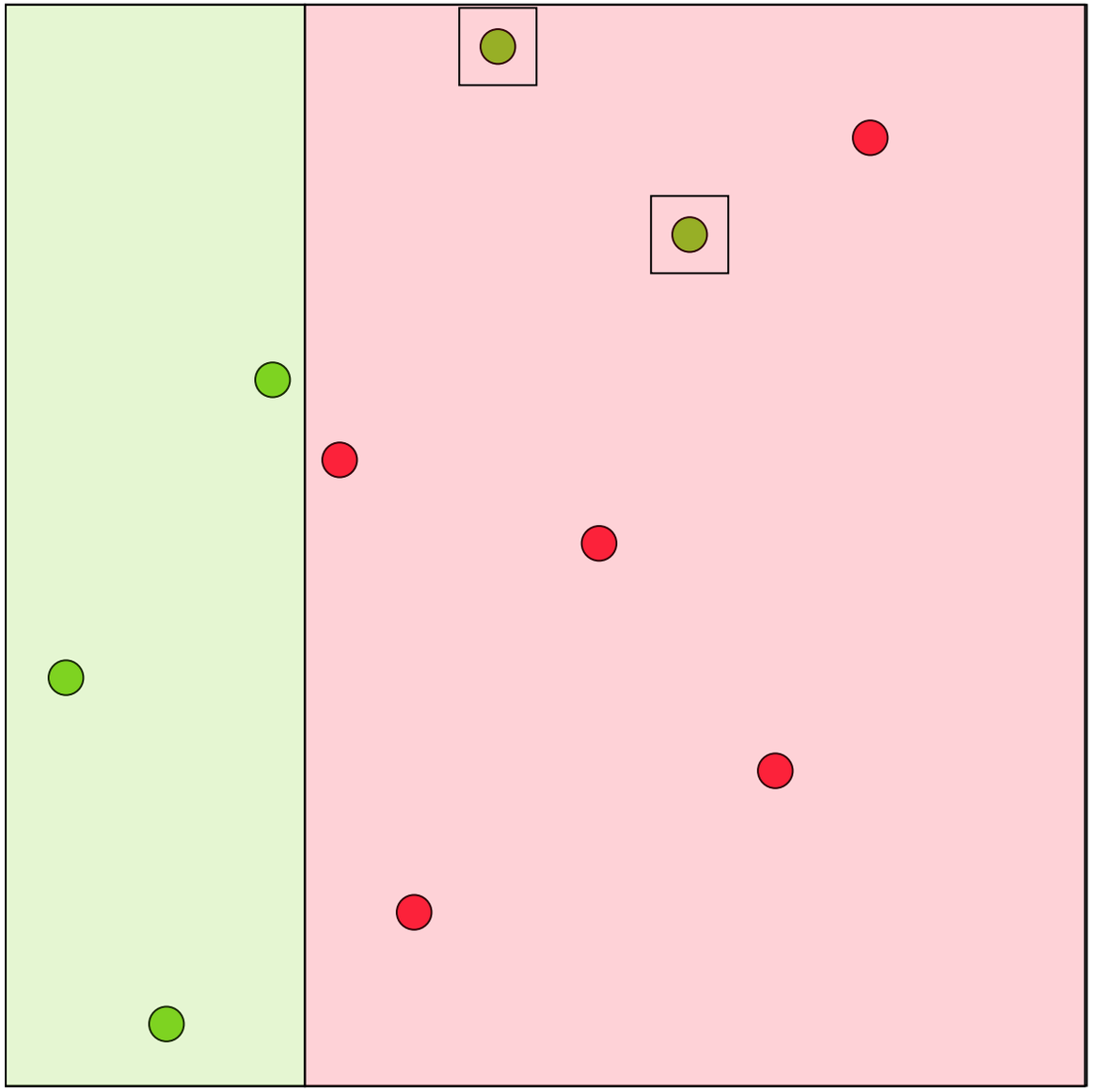


$$\widetilde{D}_1(i) = \begin{cases} 0.2, & \text{mistake} \\ 0.05, & \text{not a mistake} \end{cases}$$

$$\sum \widetilde{D}_1(i) = 0.2 \times 2 + 0.05 \times 8 = 0.8$$

AdaBoost | Round-1



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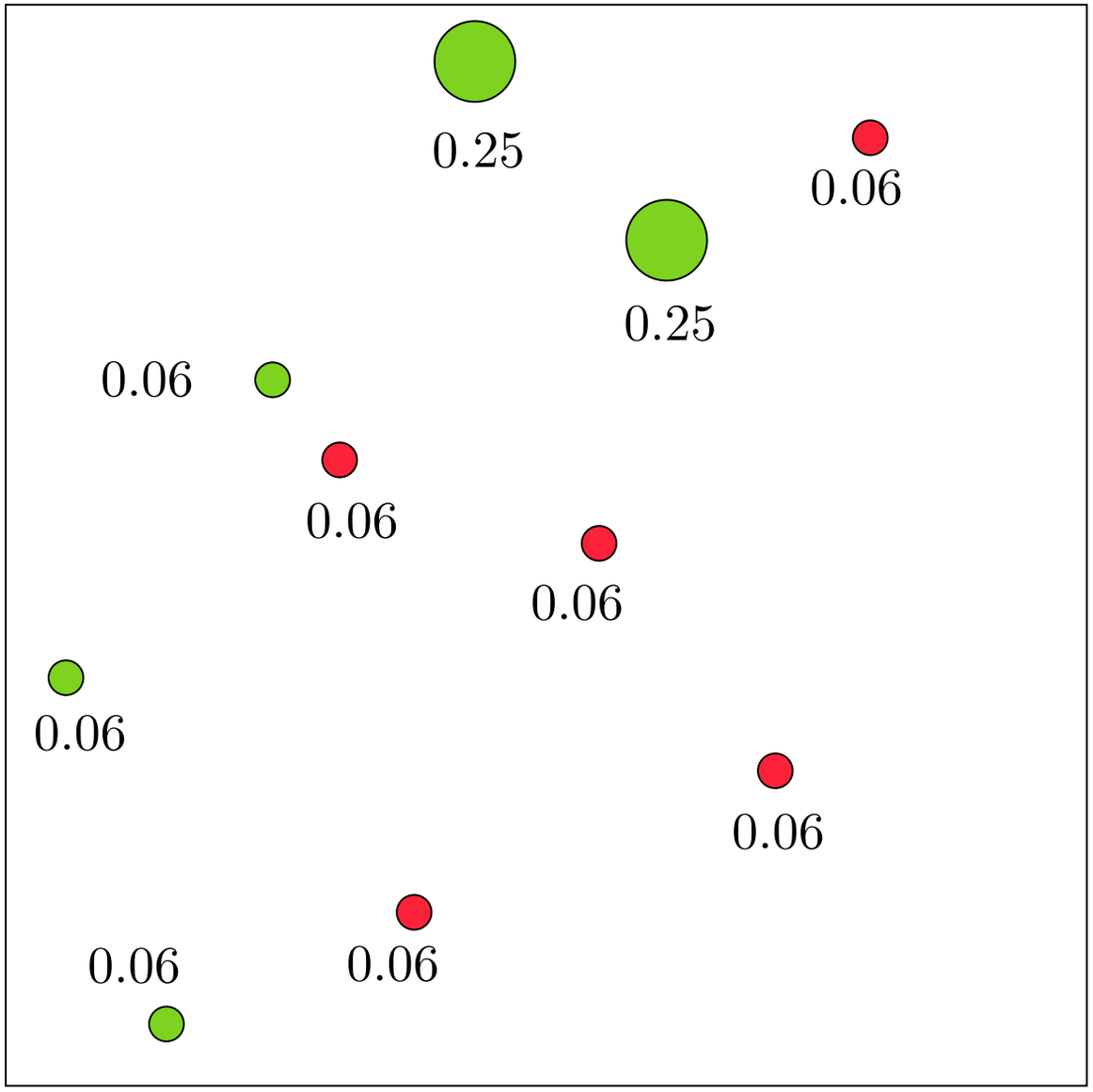
$$D_1(i) = \frac{\widetilde{D}_1(i)}{\sum \widetilde{D}_1(i)}$$

$$D_1(i) = \begin{cases} 0.25, & \text{mistake} \\ 0.06, & \text{not a mistake} \end{cases}$$

AdaBoost | Round-1

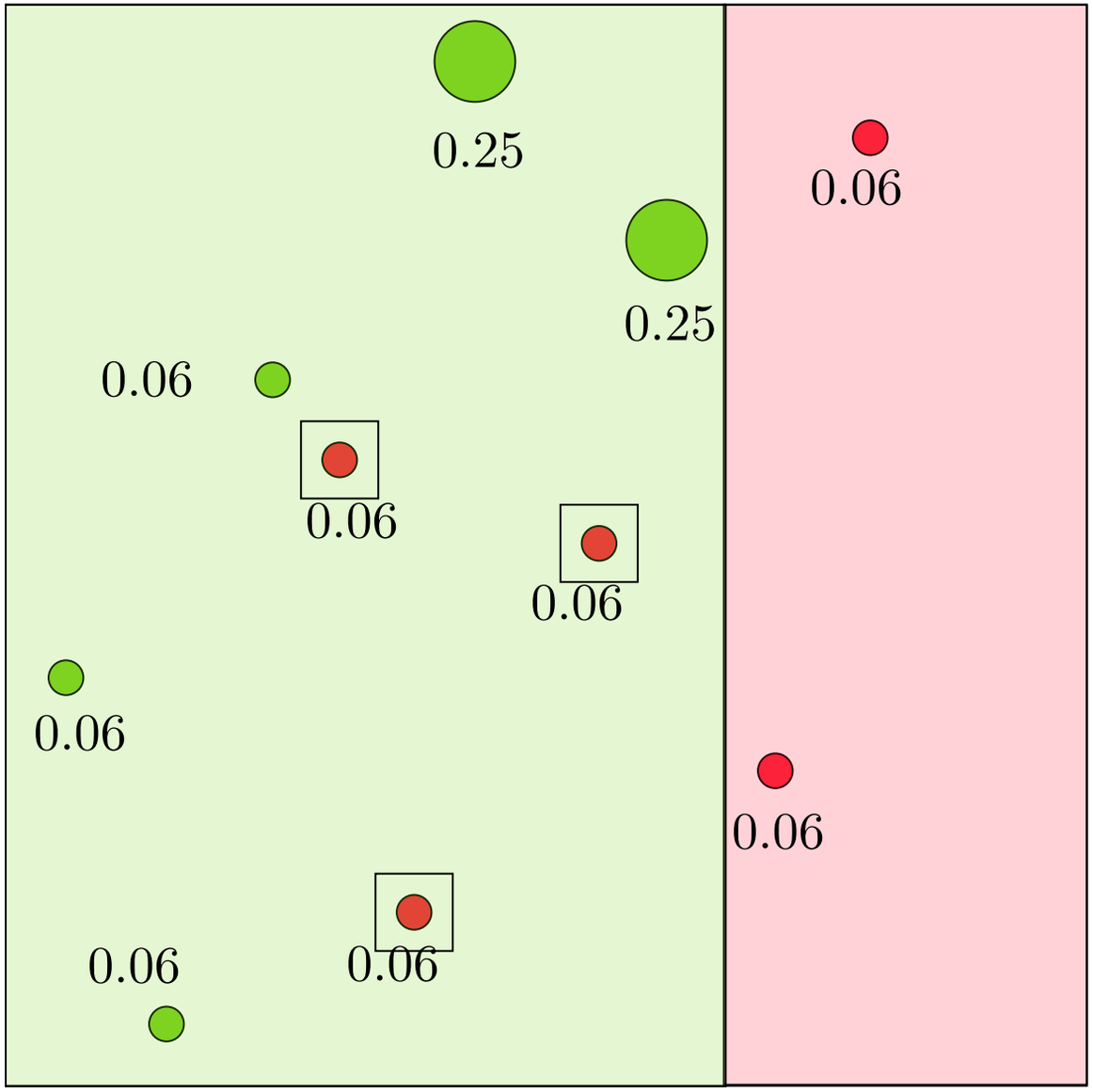
$$D_1(i) = \begin{cases} 0.25, & \text{mistake} \\ 0.0625, & \text{not a mistake} \end{cases}$$

AdaBoost | Round-2



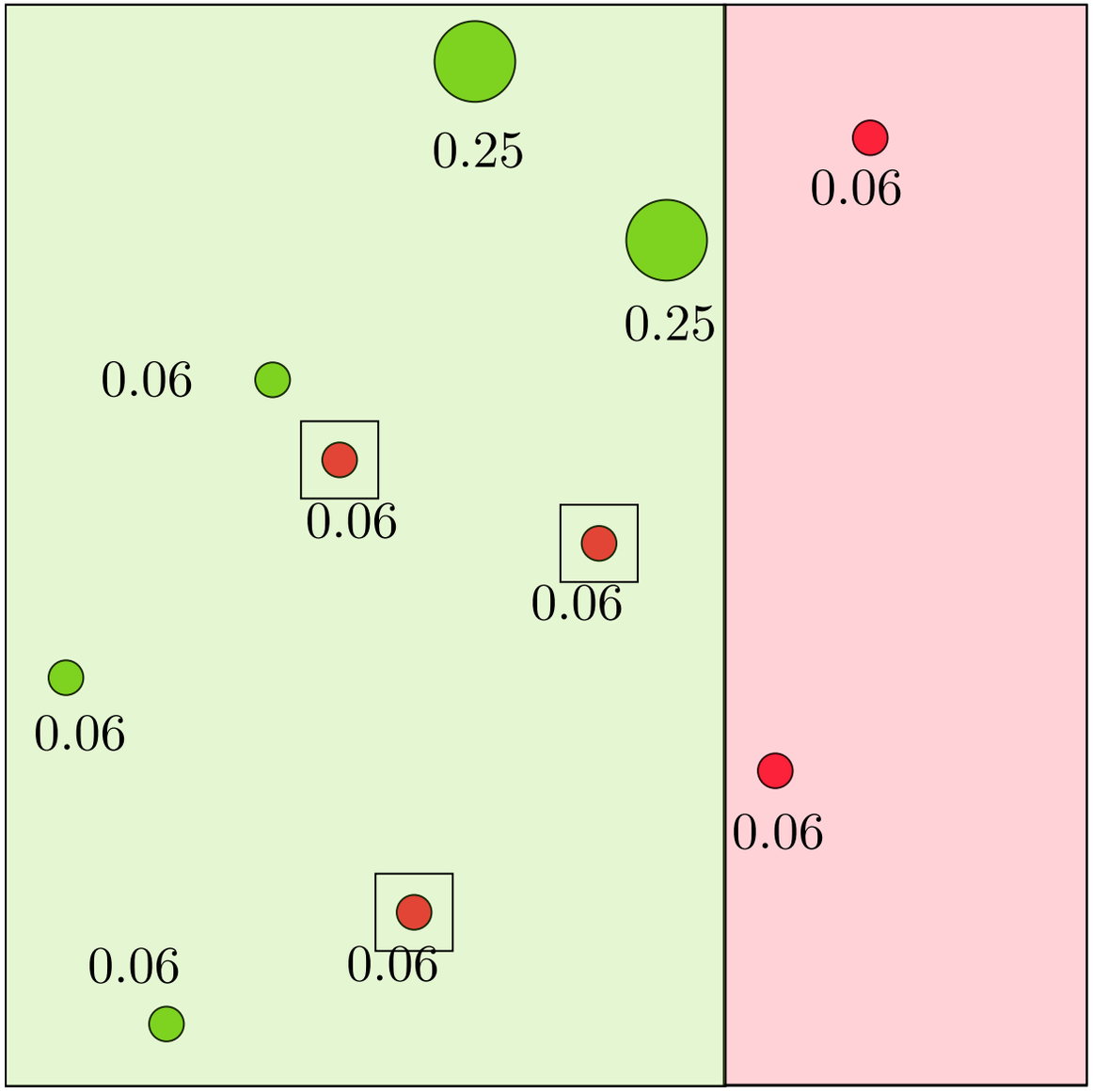
AdaBoost | Round-2

AdaBoost | Round-2



$$\epsilon_2 = 0.0625 \times 3 \approx 0.19$$

AdaBoost | Round-2

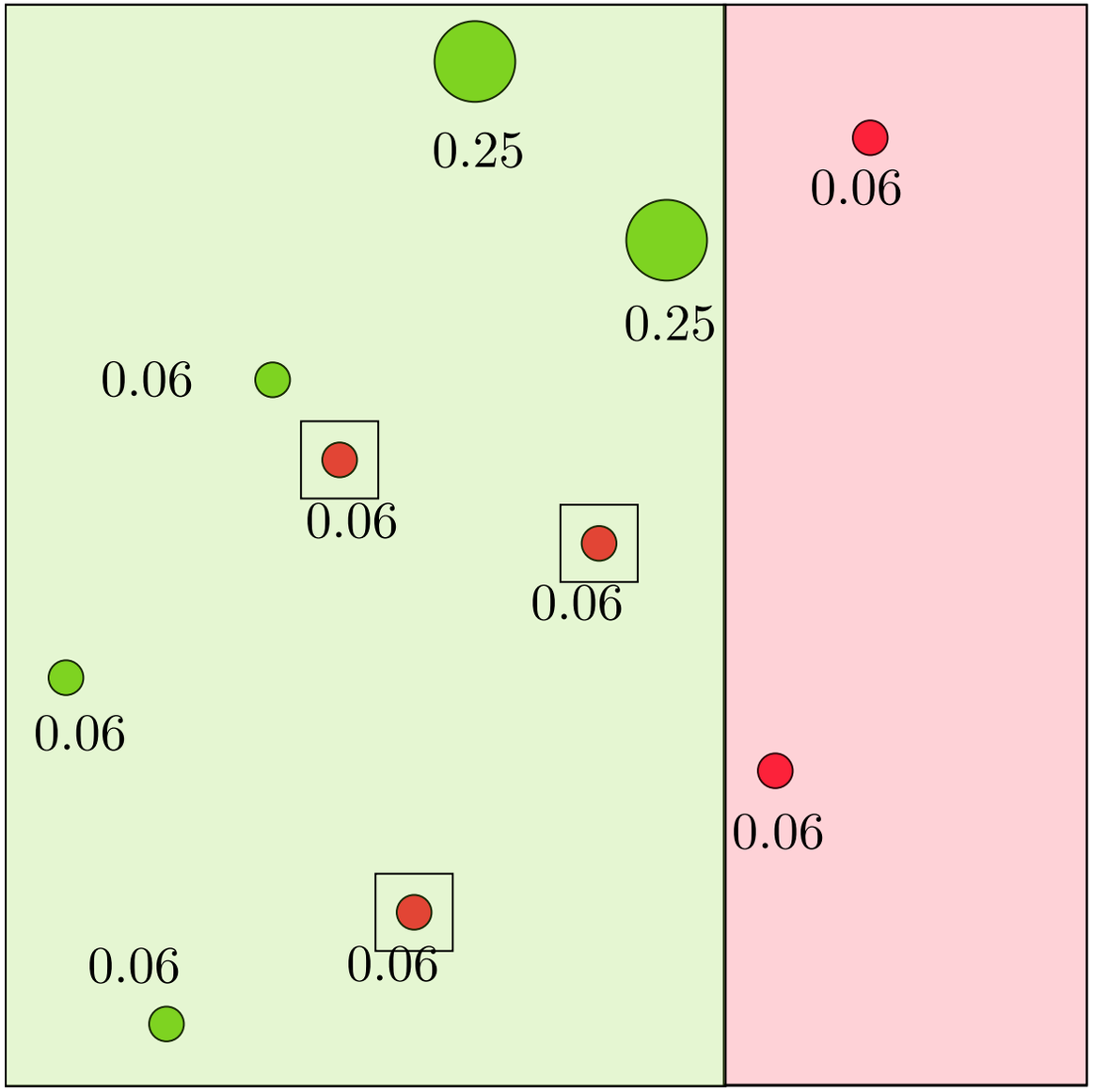


$$\alpha_2 = \frac{1}{2}\ln\left(\frac{1-\epsilon_2}{\epsilon_2}\right) = \frac{1}{2}\ln\left(\frac{1-0.19}{0.19}\right) \approx 0.73$$

$$\widetilde{D}_2(i) = \begin{cases} D_1(i)\cdot e^{\alpha_1}, & \text{mistake} \\ D_1(i)\cdot e^{-\alpha_1}, & \text{not a mistake} \end{cases}$$

$$\widetilde{D}_2(i) = \begin{cases} D_1(i)\times 2.08, & \text{mistake} \\ D_1(i)\times 0.48, & \text{not a mistake} \end{cases}$$

AdaBoost | Round-2

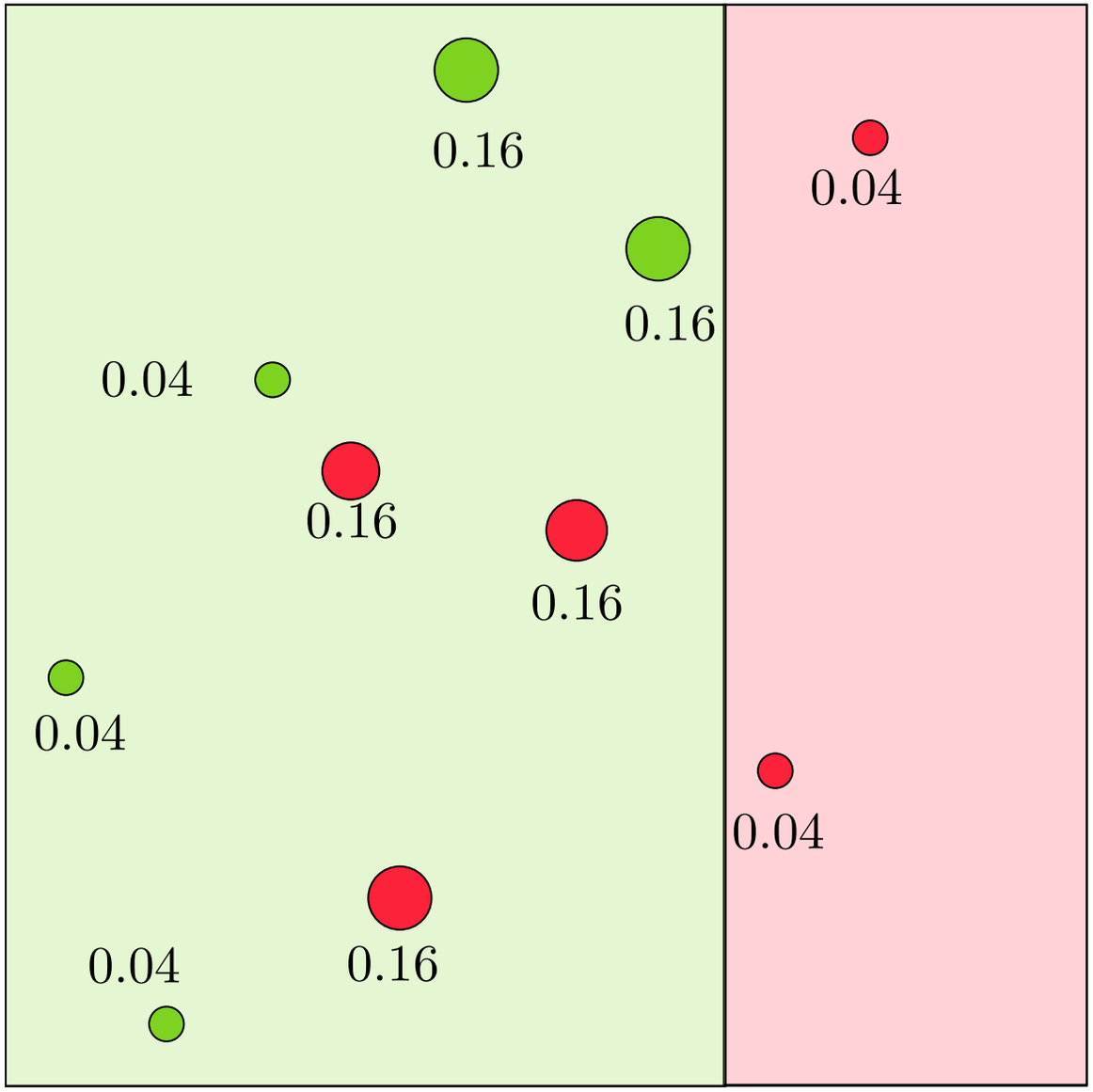


$$\widetilde{D}_2(i) = \begin{cases} D_1(i) \times 2.08, & \text{mistake} \\ D_1(i) \times 0.48, & \text{not a mistake} \end{cases}$$

$$\begin{aligned} \sum \widetilde{D}_2(i) &= 0.06 \times 2.08 \times 3 \\ &+ 0.06 \times 0.48 \times 5 \\ &+ 0.25 \times 0.48 \times 2 \quad = \quad 0.76 \end{aligned}$$

$$D_2(i) = \frac{\widetilde{D}_2(i)}{\sum \widetilde{D}_2(i)}$$

AdaBoost | Round-2



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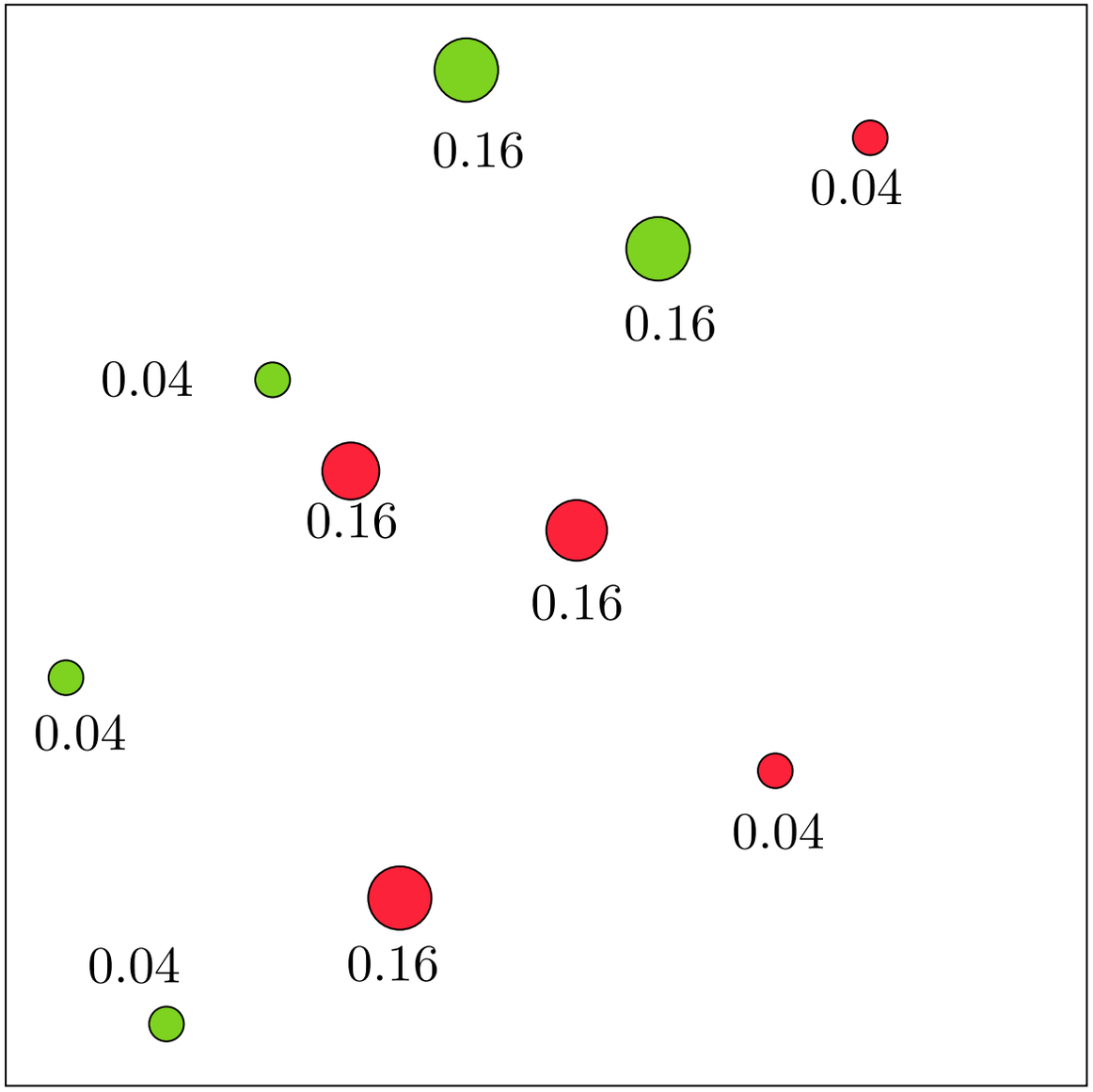
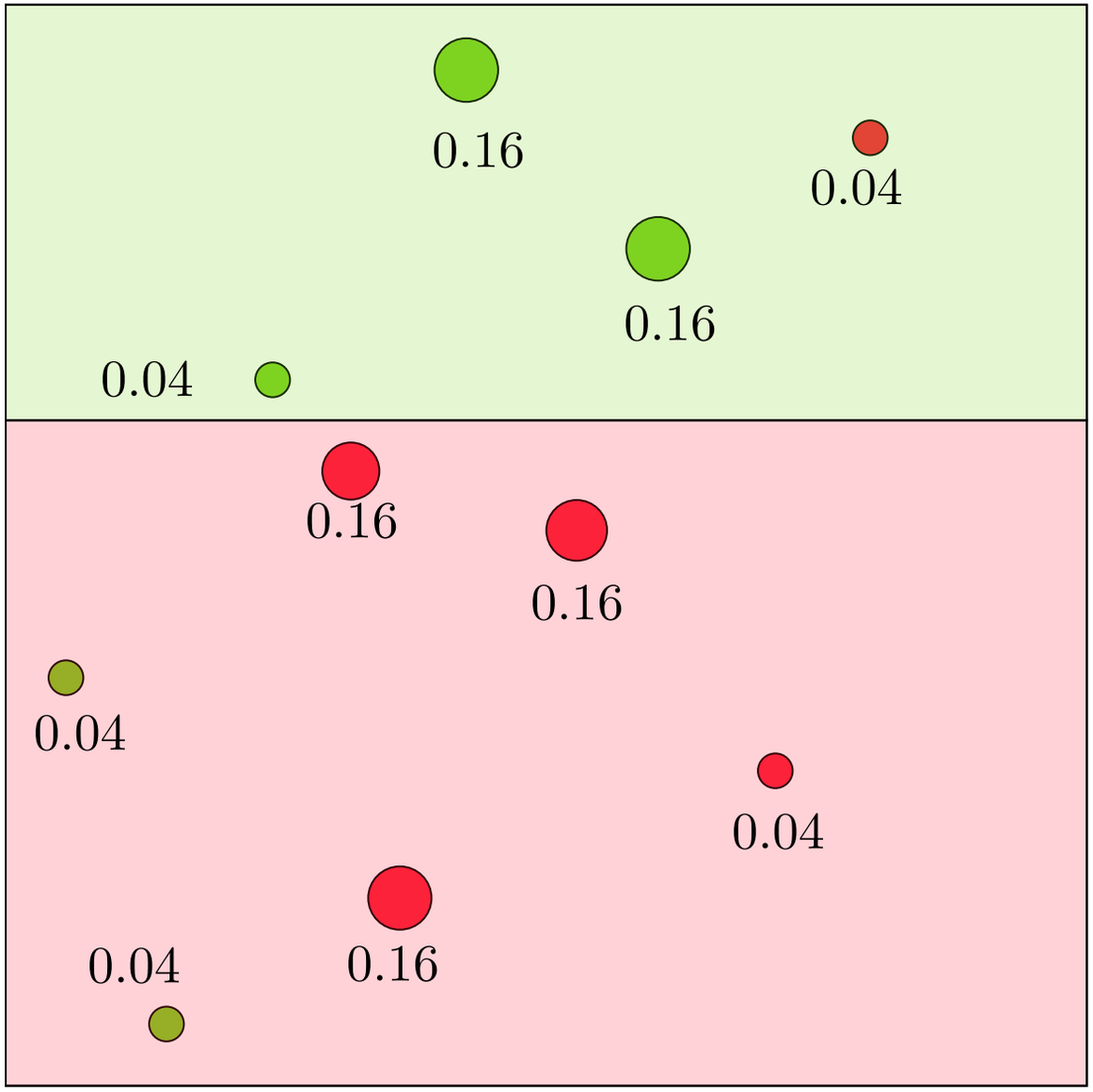
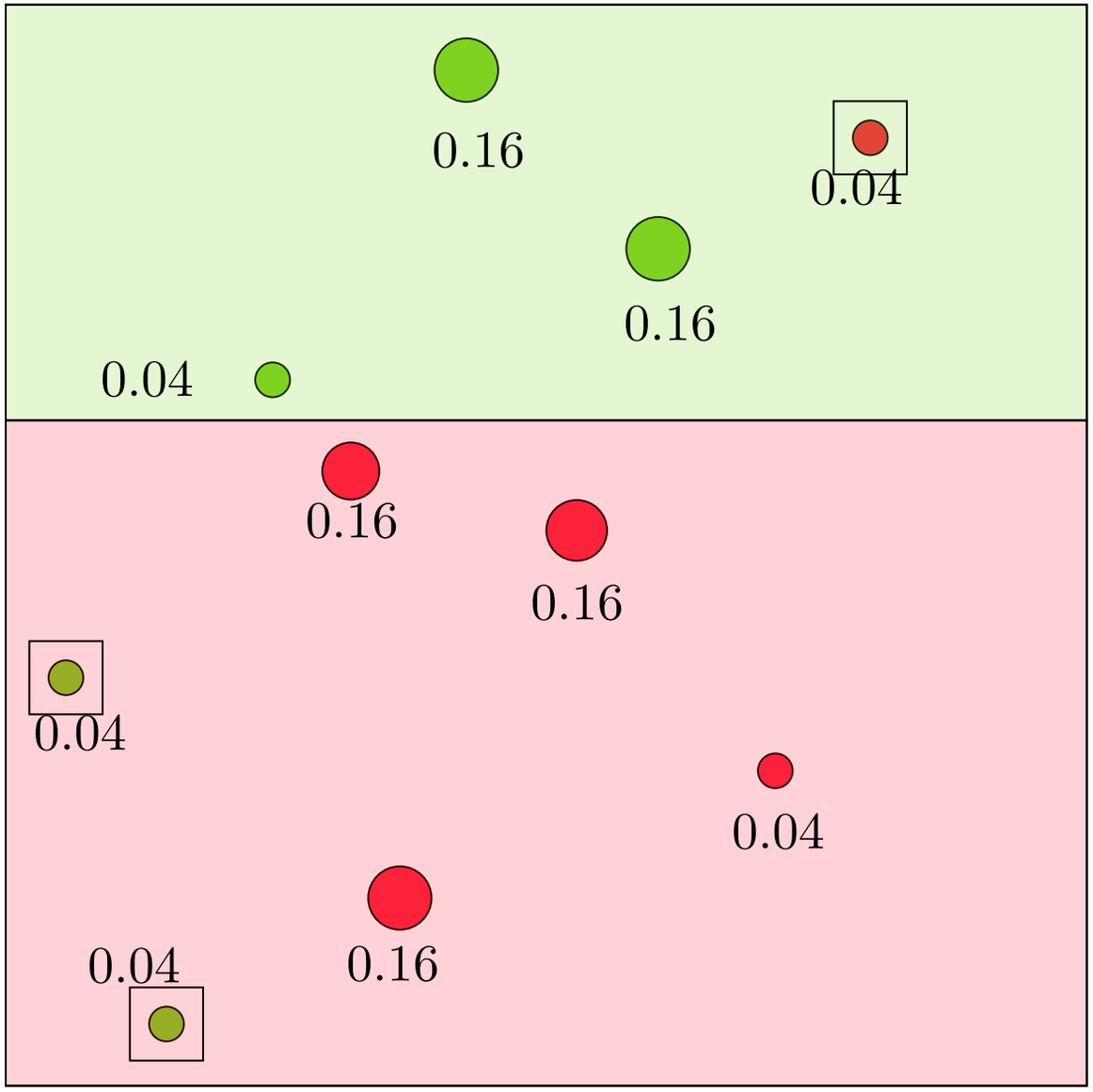
AdaBoost | Round-3



AdaBoost | Round-3

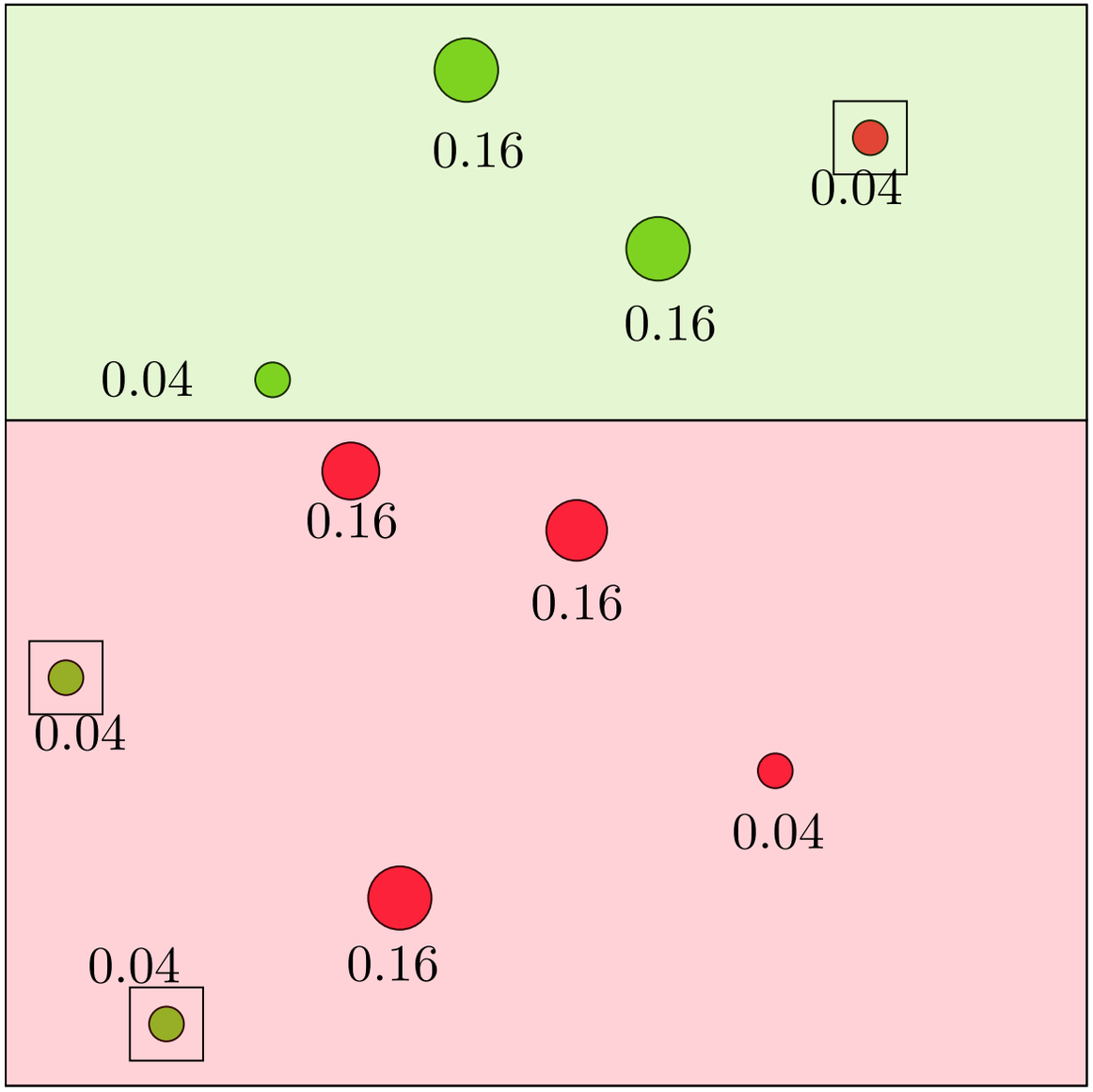


AdaBoost | Round-3



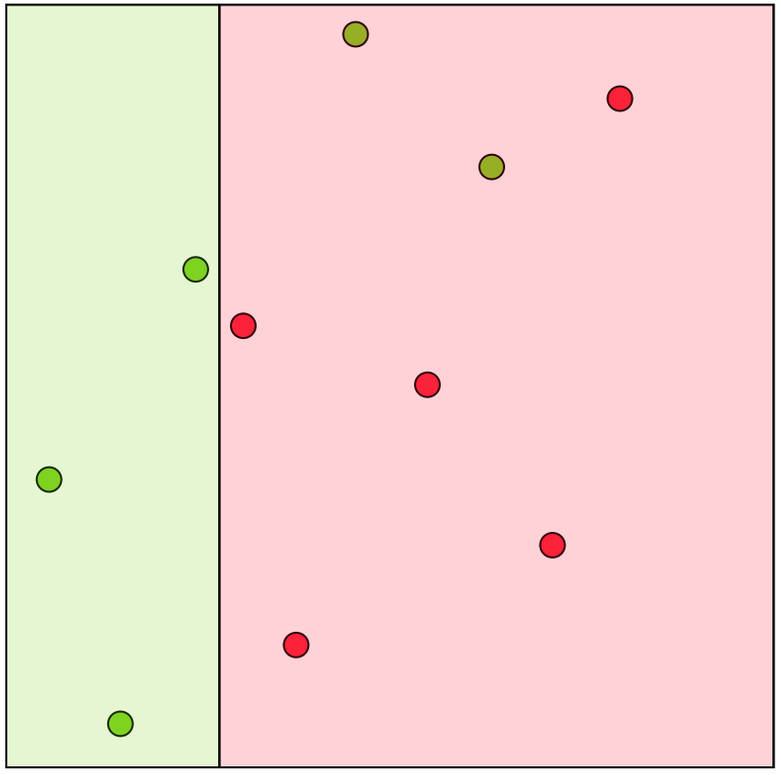
$$\epsilon_3 = 0.04 \times 3 = 0.12$$

AdaBoost | Round-3

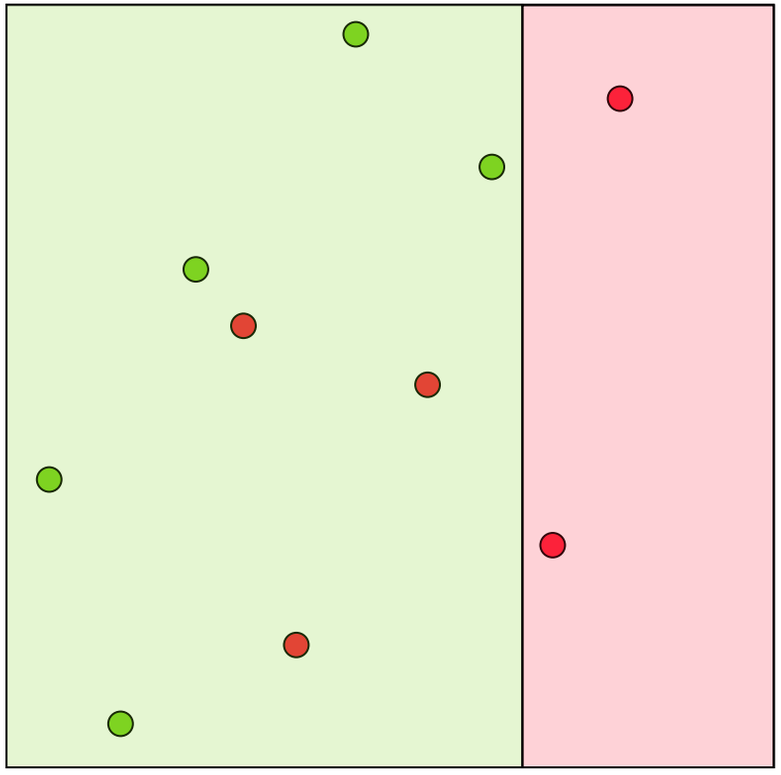


$$\alpha_3 = \frac{1}{2}\ln\left(\frac{1-\epsilon_3}{\epsilon_3}\right) = \frac{1}{2}\ln\left(\frac{1-0.12}{0.12}\right) \approx 1$$

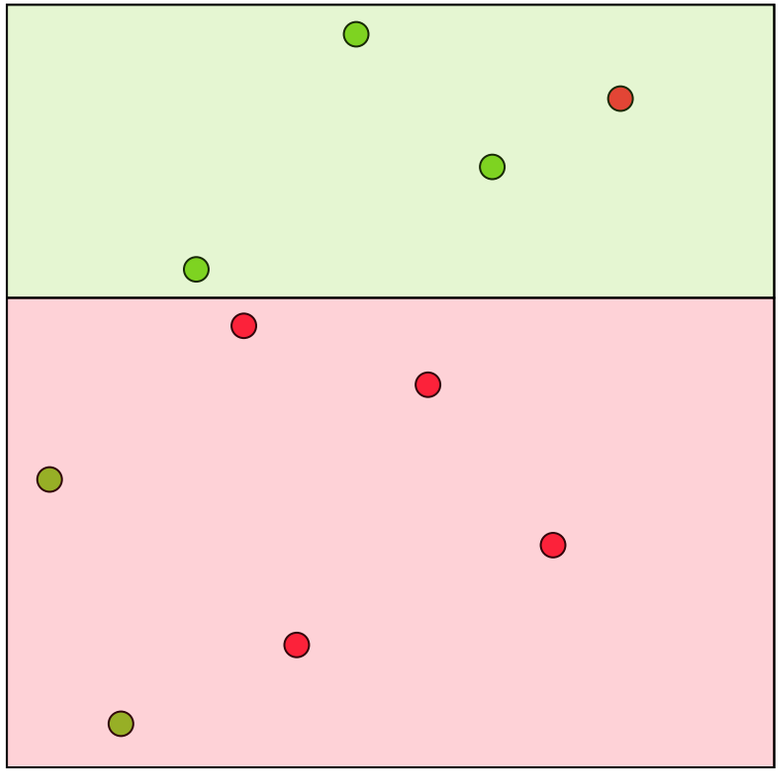
AdaBoost | Ensemble



$\alpha_1 \approx 0.69$

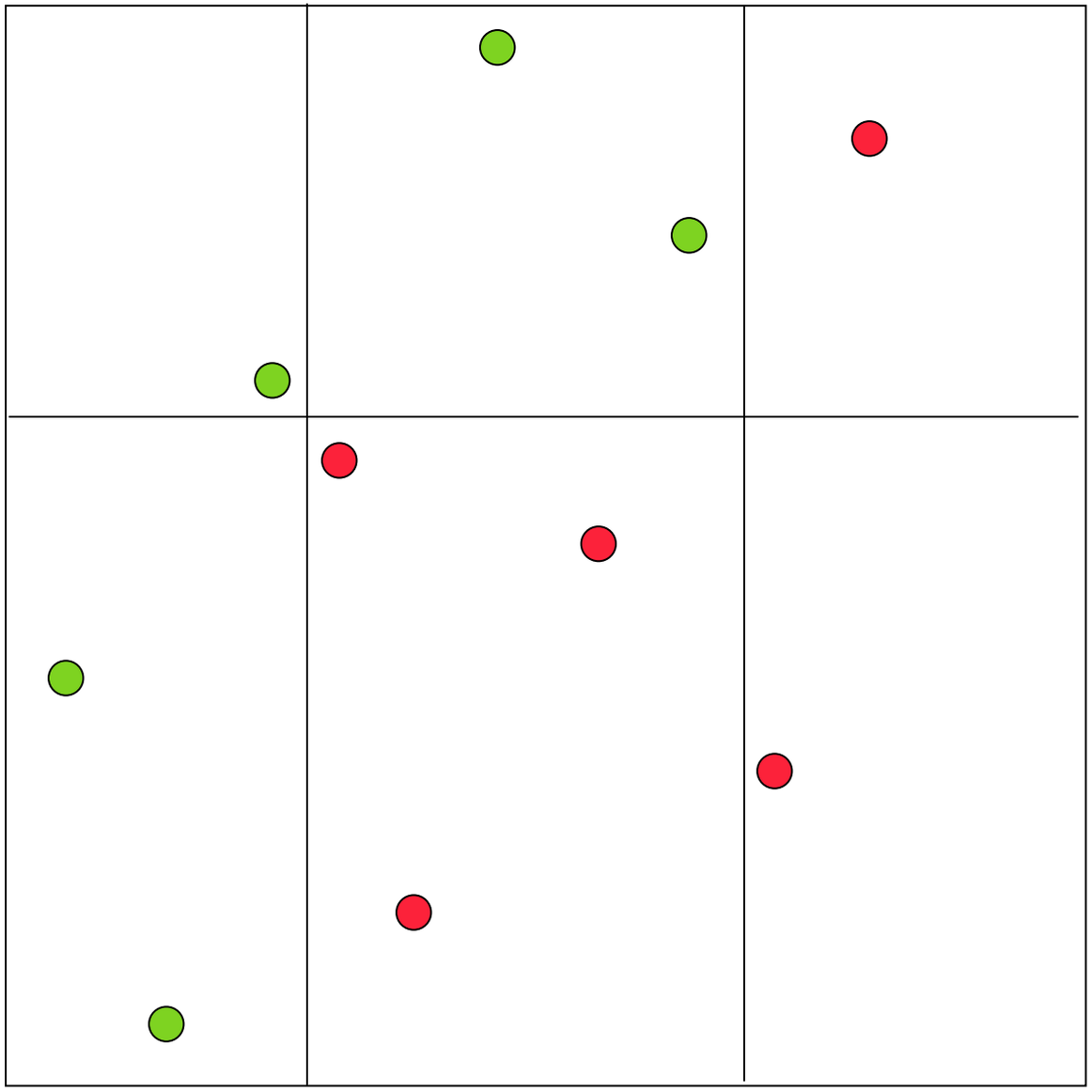


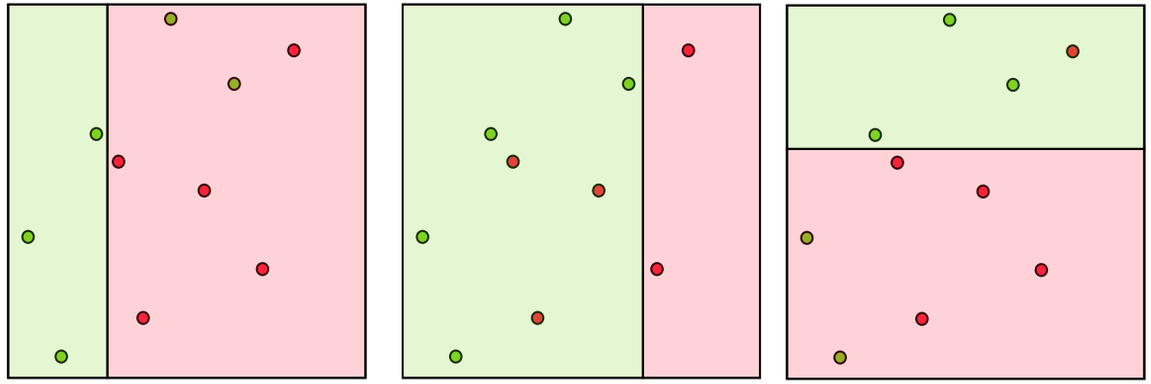
$\alpha_2 \approx 0.73$



$\alpha_1 \approx 1$

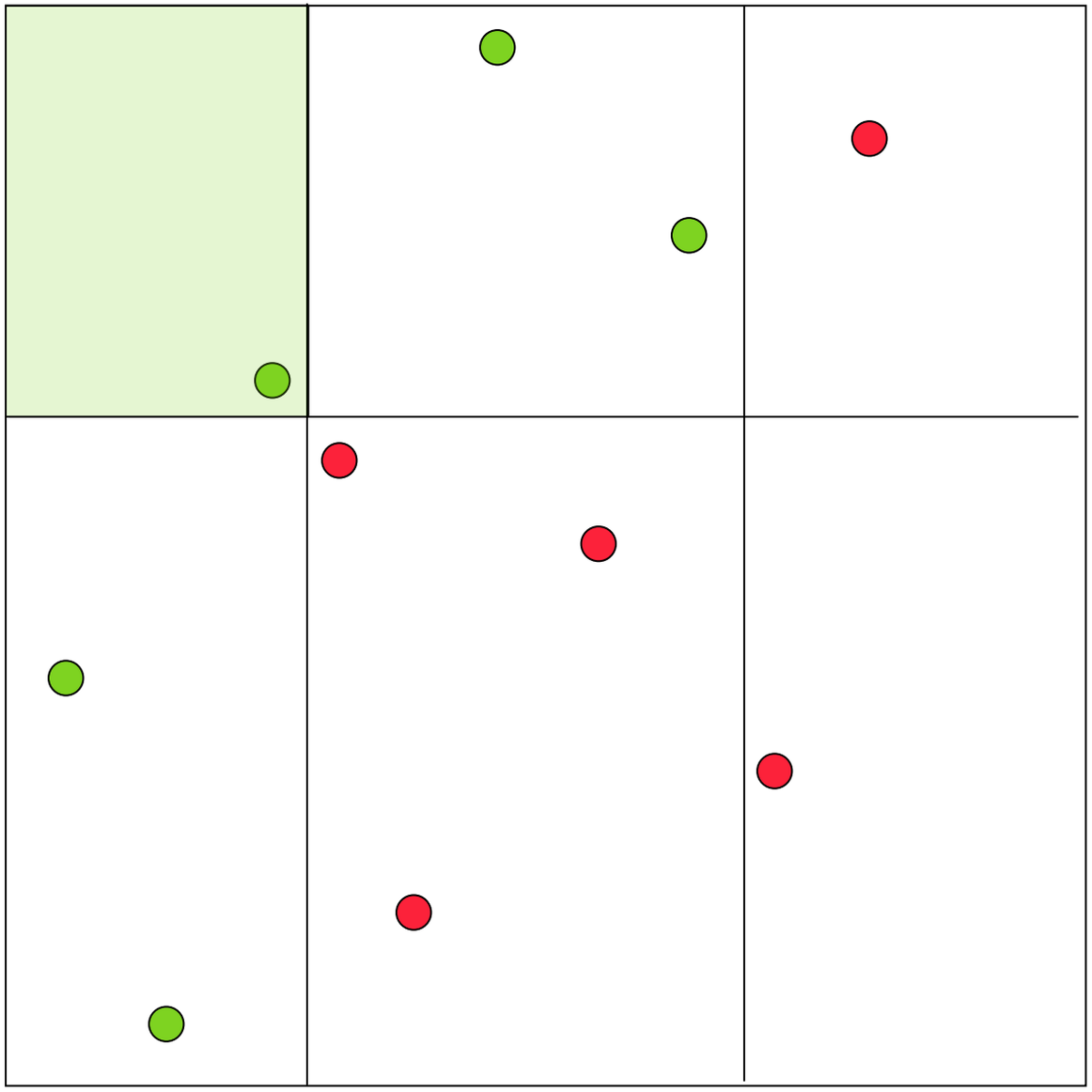
AdaBoost | Final Classifier | Decision Regions

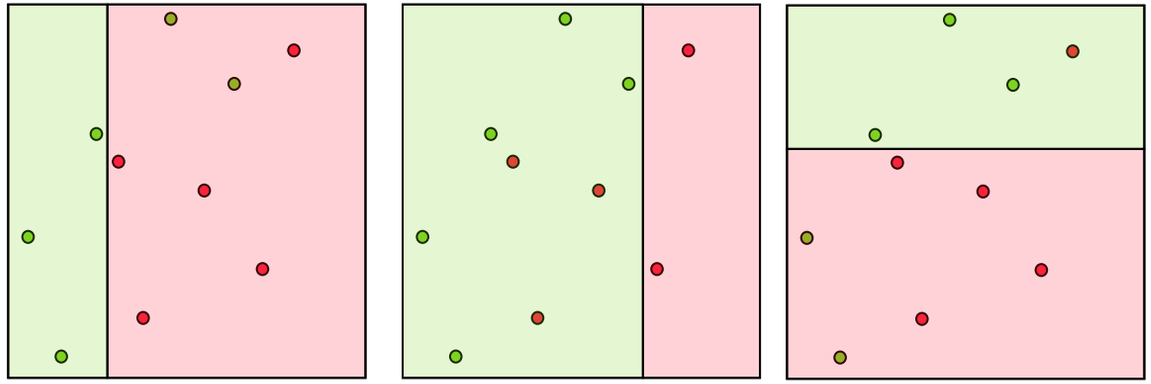




$\alpha_1 \approx 0.69$ $\alpha_2 \approx 0.73$ $\alpha_1 \approx 1$

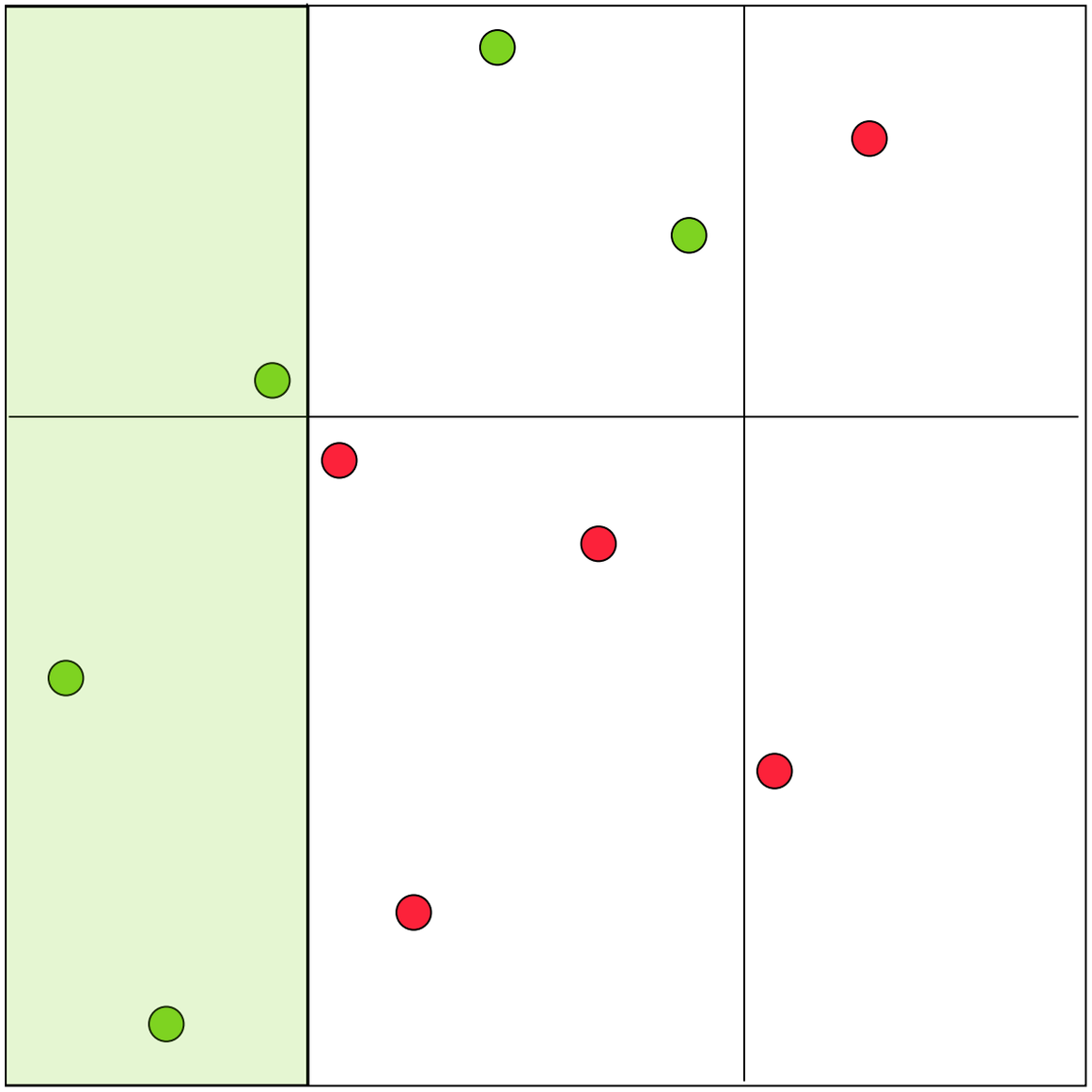
AdaBoost | Final Classifier | Decision Regions

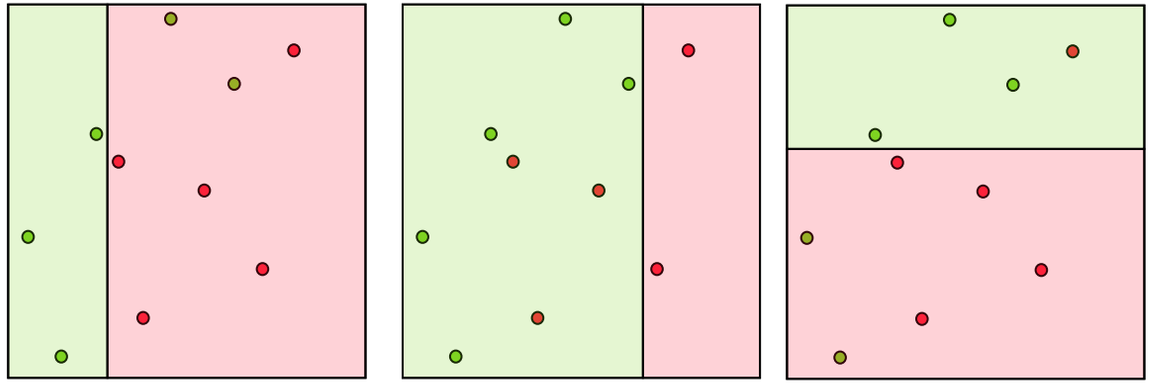




$\alpha_1 \approx 0.69$ $\alpha_2 \approx 0.73$ $\alpha_1 \approx 1$

AdaBoost | Final Classifier | Decision Regions





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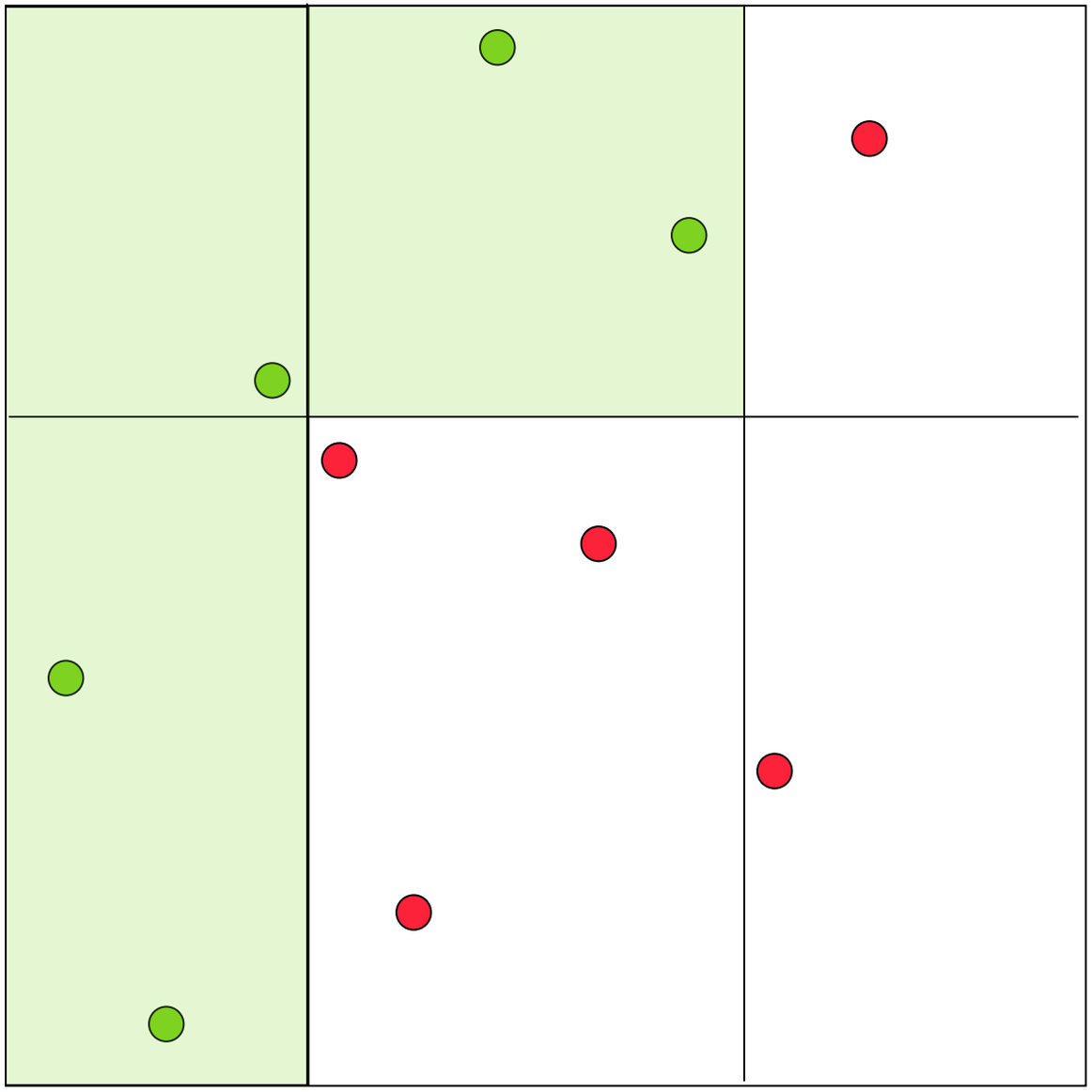

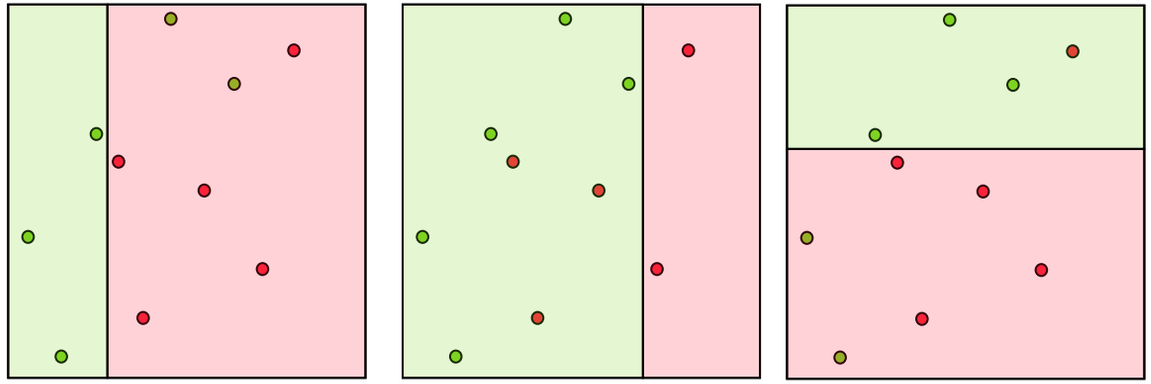

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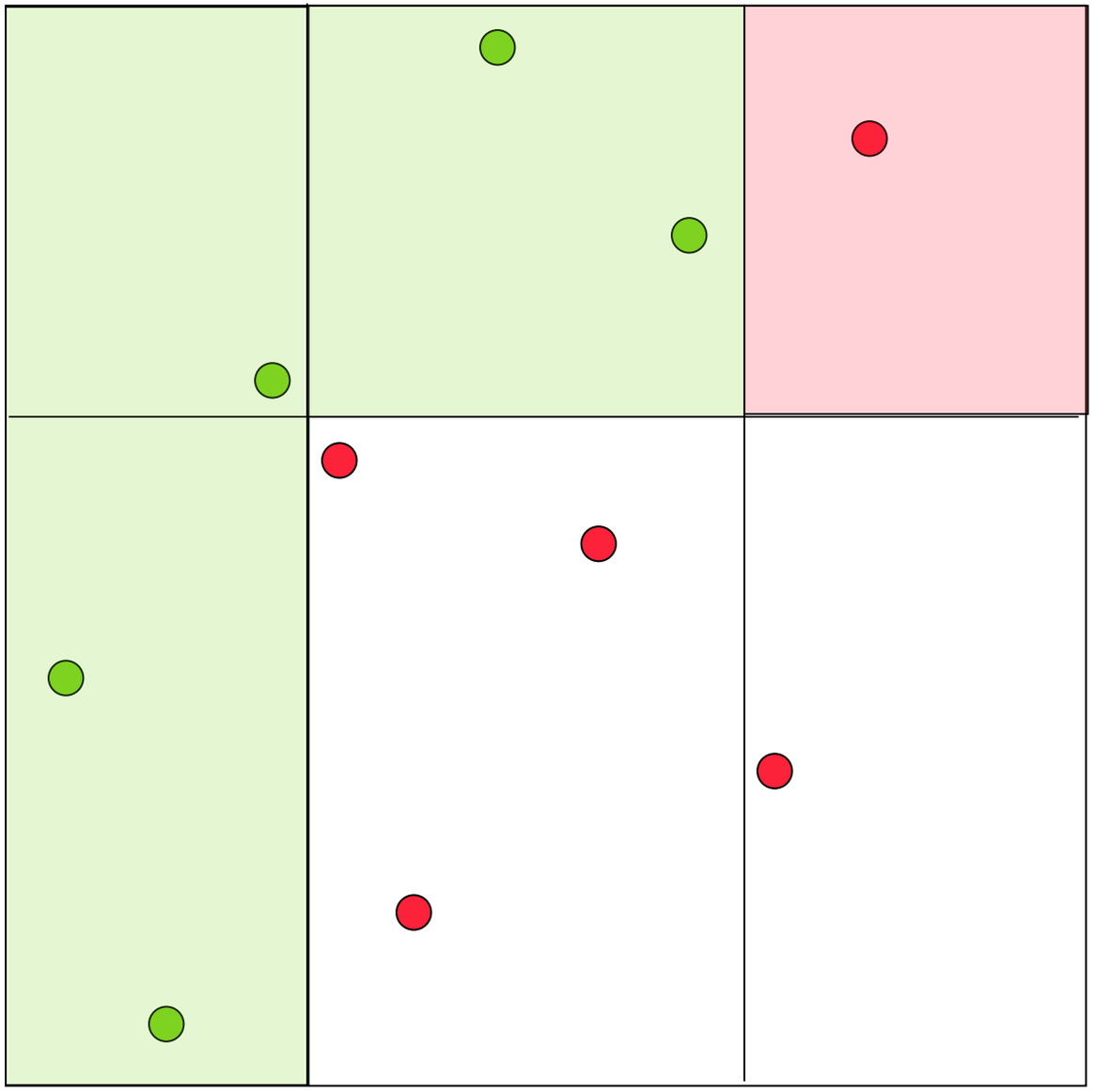

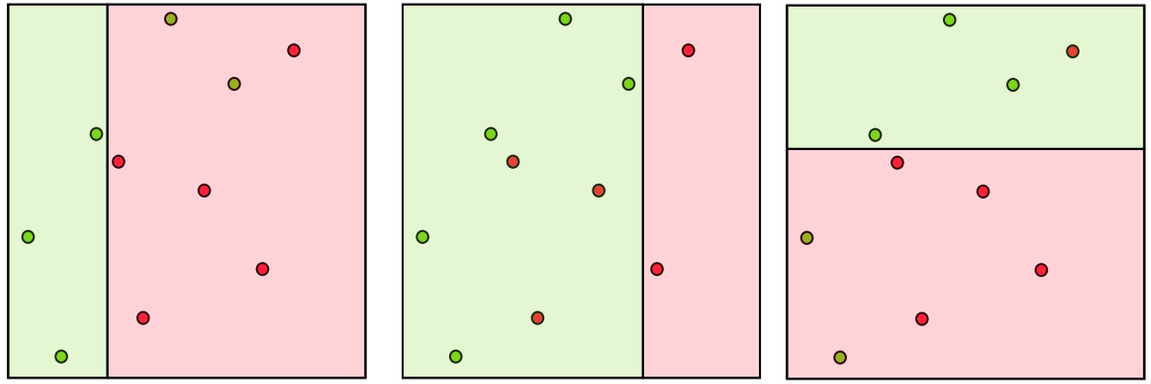

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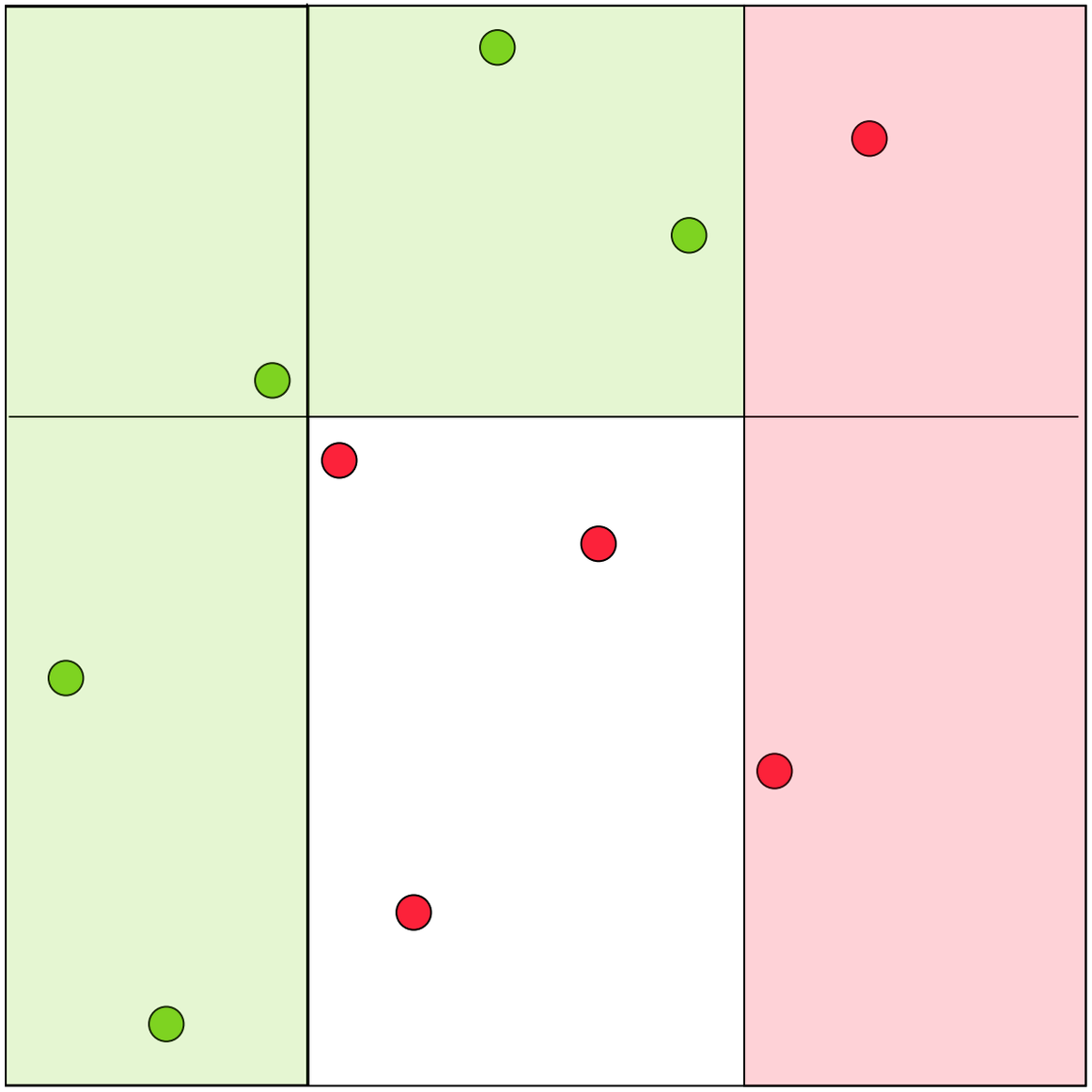

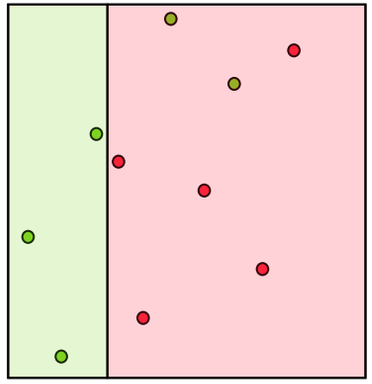

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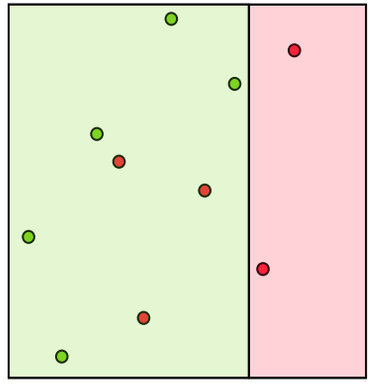

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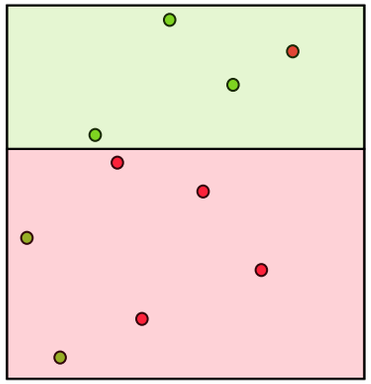

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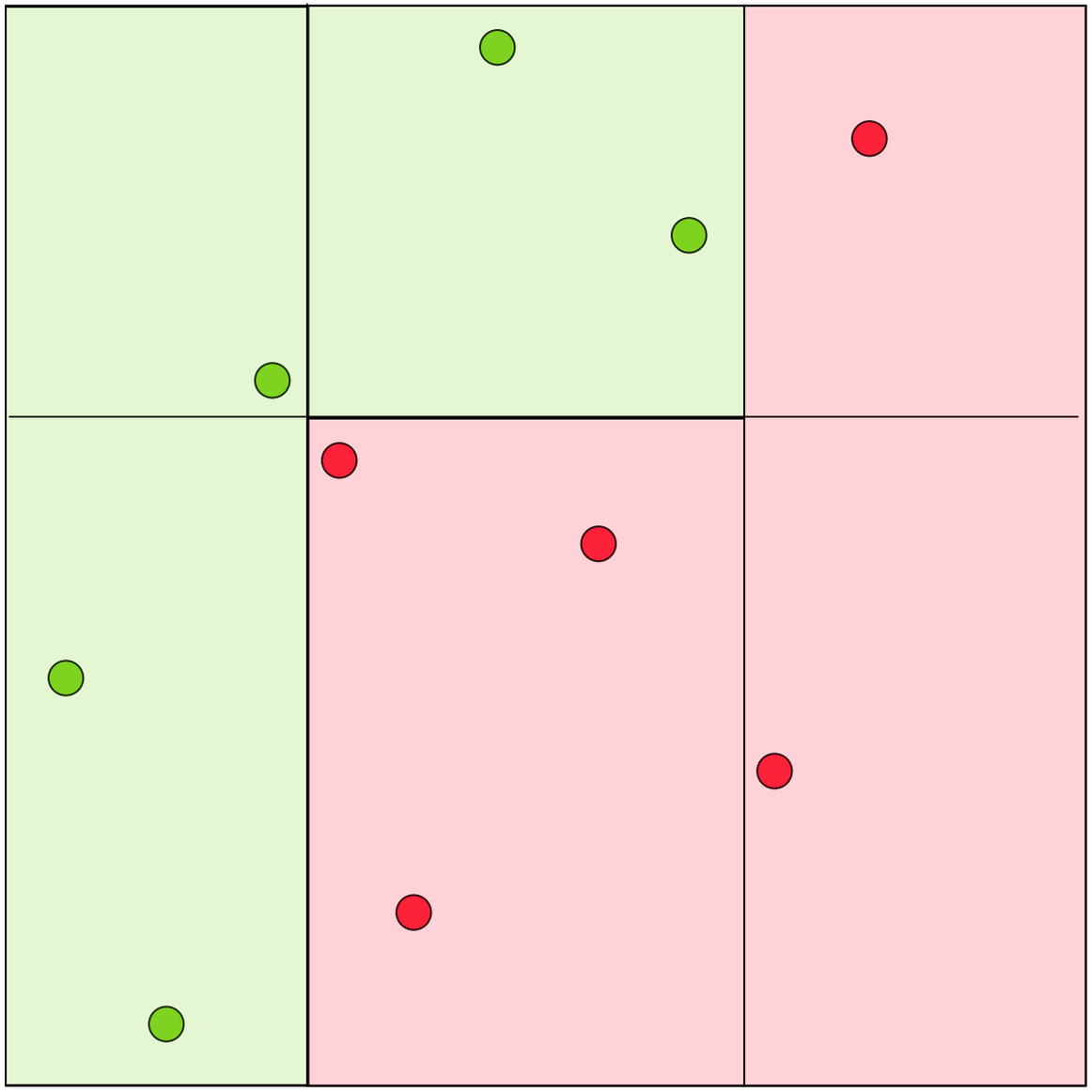
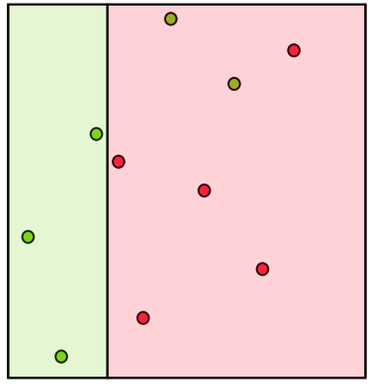
AdaBoost | Final Classifier | Decision Regions





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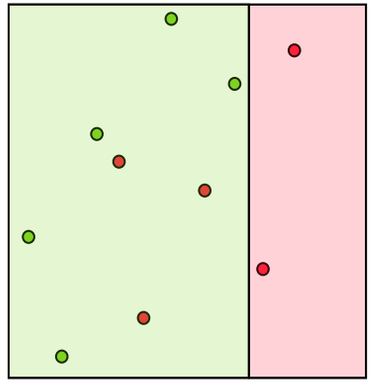

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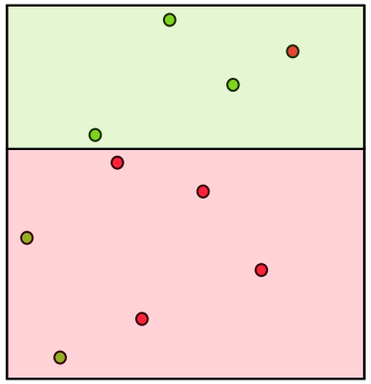

$\alpha_1 \approx 1$

References

- This example has been entirely borrowed from Kasper Green's YouTube video.
- The exact configuration worked out in the video has been reproduced here.
- No screenshots have been used and the figures have been reconstructed from scratch.